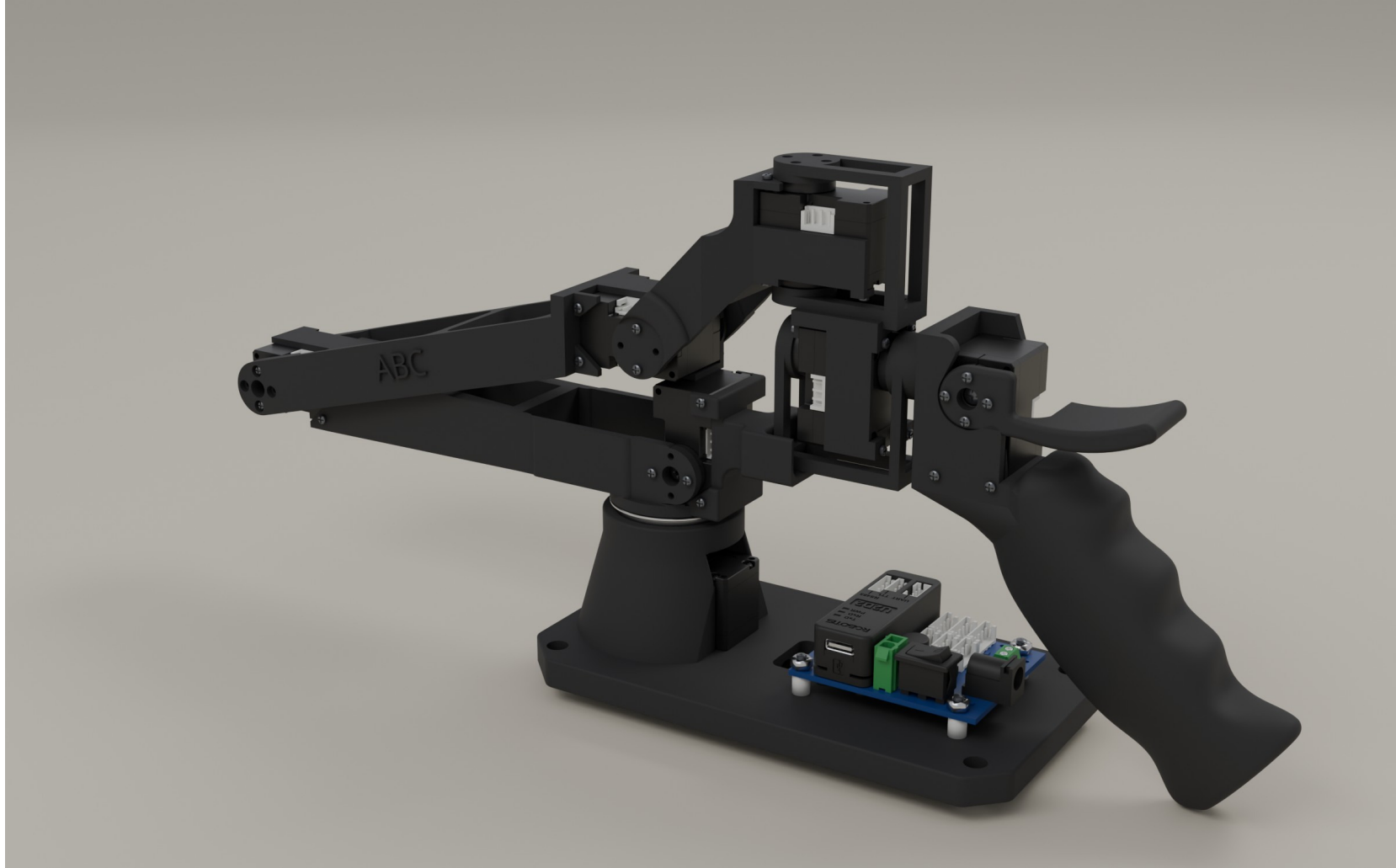


ABC GELLO



ASSEMBLY

BOM

Item	Qty	Unit	Total	Link
DYNAMIXEL XL330-M077-T	7	\$27.49	\$192.43	Buy
FPX330-H101 set (wheels + caps)	2	\$10.78	\$21.56	Buy
U2D2	1	\$35.31	\$35.31	Buy
Bearings	1	\$12.27	\$12.27	Buy
M2/M3 screw kit	1	\$14.99	\$14.99	Buy
U2D2 Power Hub Board set	2	\$21.85	\$43.70	Buy
5V power supply	2	\$7.99	\$15.98	Buy

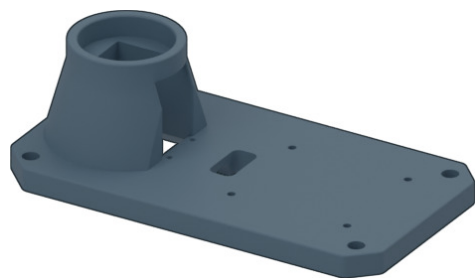
Total \$336.24

Recommended tool: electric screwdriver [View on Amazon ↗](#)

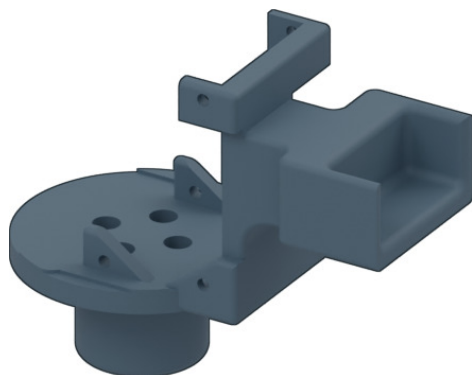
Quoted prices; shipping and tax excluded. Use the Buy links to open suppliers.

M3 × 10 mm screws: included in the BOM screw kit. Spacers: included with the power hub.

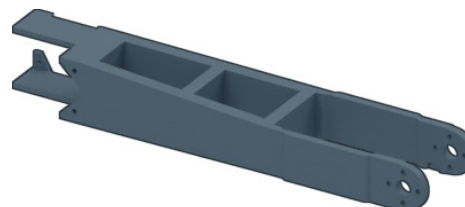
Parts



1× 0.stl
Base



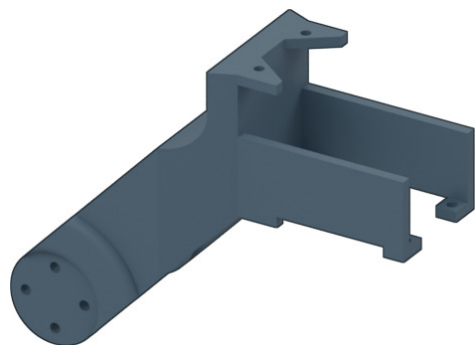
1× 1.stl
Shoulder



1× 2.stl
Upper arm



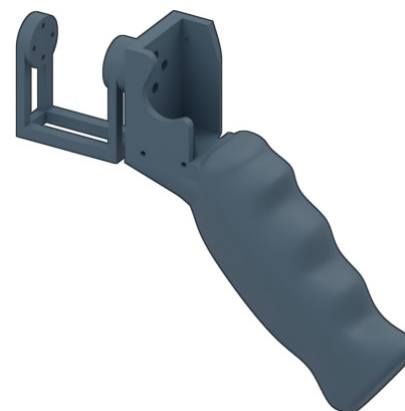
1× 3.stl
Forearm / ABC



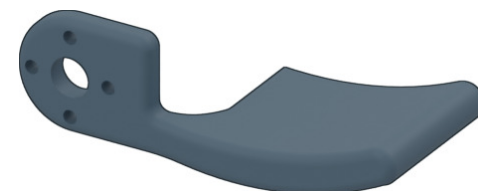
1× 4.stl
Wrist fork



1× 5.stl
Wrist cage



1× 6.stl
Handle



1× 7.stl
Trigger

Blue = part being added. Grey = previous assembly.

Position the part first, then follow the screw close-ups. Front and rear faces have separate counts.

3D printing notes

Print every part with supports enabled.

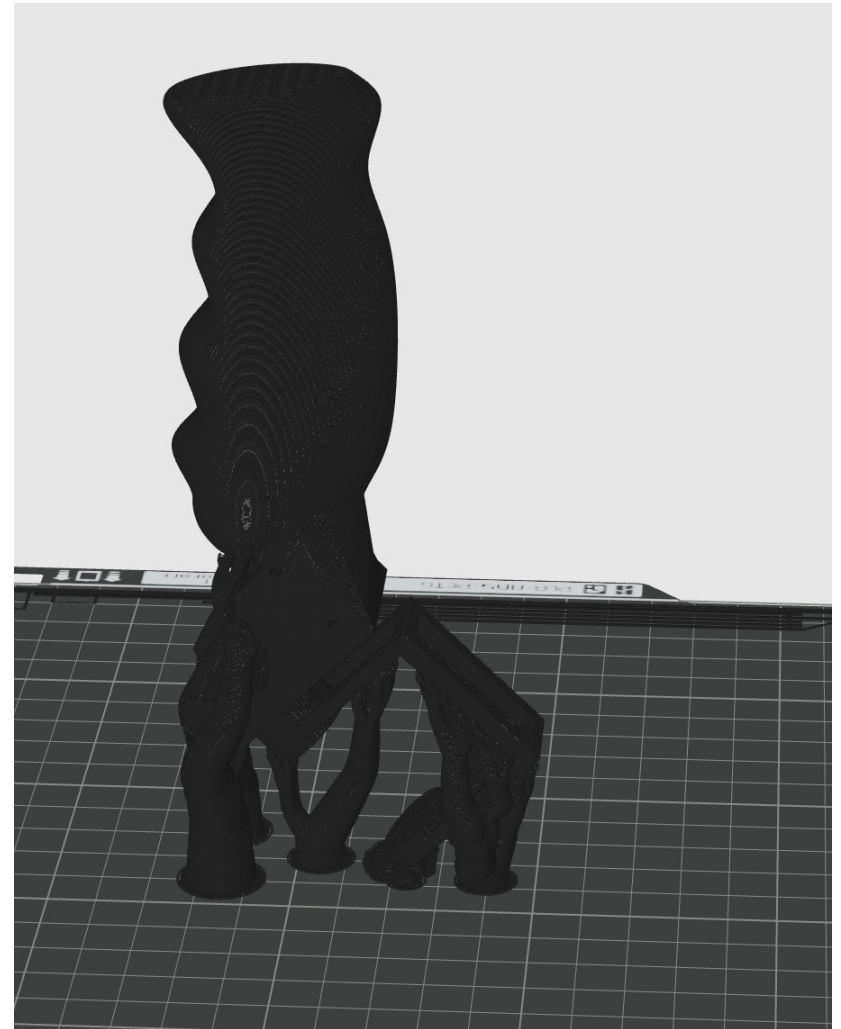
Handle link: 6.stl

Print upside down with supports for best results, as shown.

Grip points upward; the mounting fork sits near the build plate.

Use this orientation for the compact handle variant too.

Remove supports from mounting holes and motor pockets before starting assembly.



Builder's slicer example: handle upside down with supports

Before assembly

1

Finish the mechanical assembly first.

Wiring starts at step 47, after the power hub is secured.

2

Label the seven physical joints M0–M6.

These labels do not prescribe numeric bus IDs.

3

Complete wheel → cap → screw on M1–M5.

M0 and M6 use the existing front horn only.



Wheel



Cap



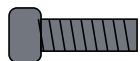
Screw

Leave the cables disconnected during these assembly steps.

The final section shows the separate base lead and the M1–M6 chain.

Screw heads are schematic; check that the supplied heads fit the recesses.

Screws



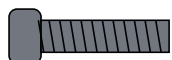
M2 × 6 TAP

Front horn + rear wheel mounting

28 used · 42 supplied



1:1 at 100% print



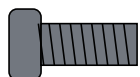
M2 × 8 TAP

Motor body → printed holder

38 used · 70 supplied



1:1 at 100% print



M2.6 × 6 TAP

Rear idler retention

5 used · 16 supplied



1:1 at 100% print

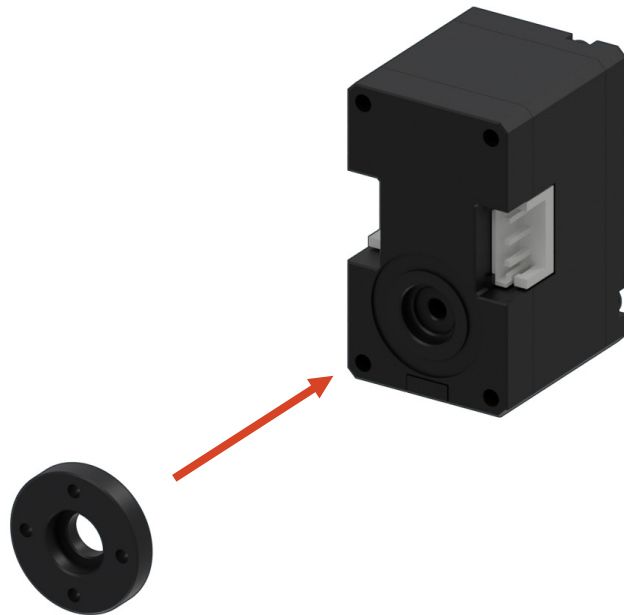
TAP = tapping screw. Two-sided horn/wheel joints use two opposite screws per face, rotated 90° between faces.

The accessory pack also includes M2 × 4 screws; keep those separate.

Hub: 4× M3 × 10 mm screws from the BOM screw kit. Use the spacers included with the power hub.



1



5 motors

1 part per motor

Place the wheel on the rear boss of M1.

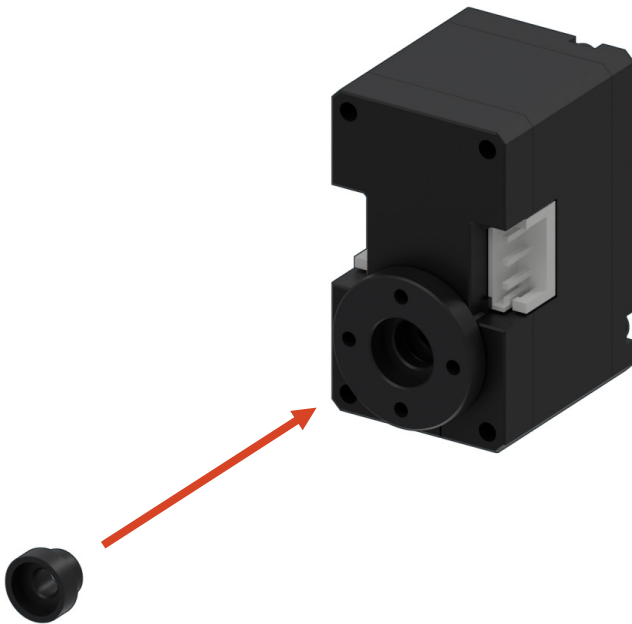
Repeat this three-step stack for M1-M5; keep the front horn fitted.



5 motors

1 part per motor

2



Seat the small cap in the centre of the rear wheel.

The cap goes between the wheel and the screw head.



1× retention screw



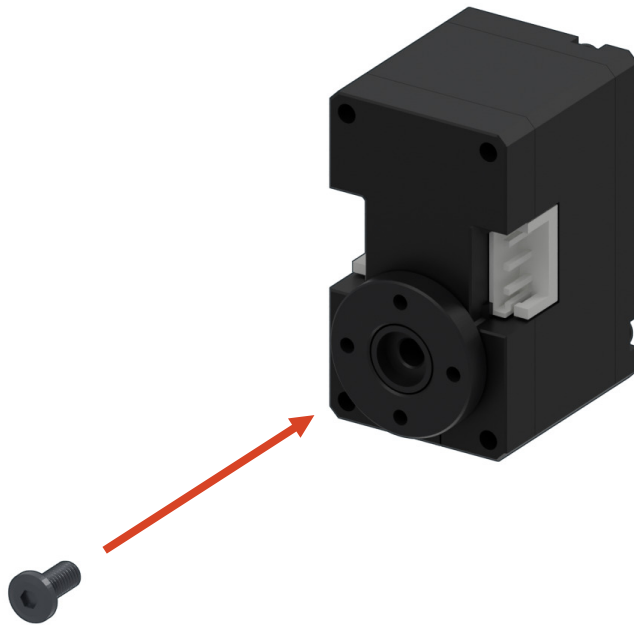
1× M2.6 × 6 TAP

Length under head, mm

5 motors

1 part per motor

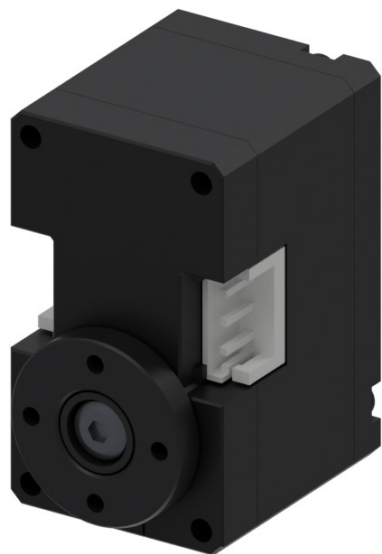
3



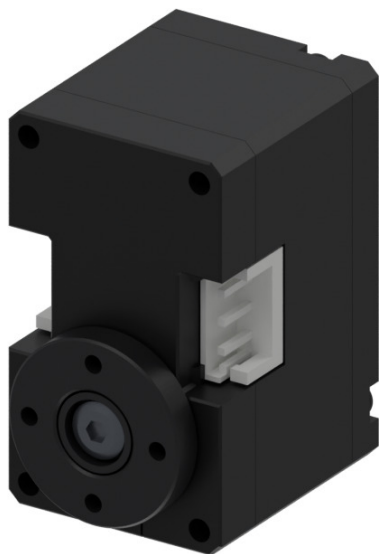
Insert the retention screw through the cap into the motor.
Check that the rear wheel turns freely. M0 and M6 get no rear wheel.

Five completed motors

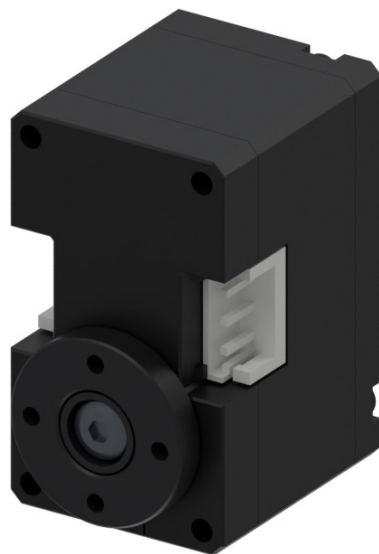
Keep each wheel, cap and retention screw fitted for all later steps.



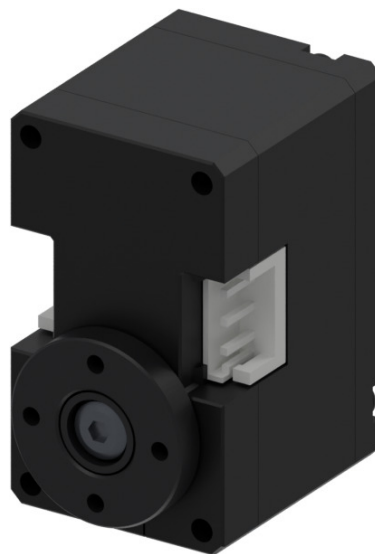
M1



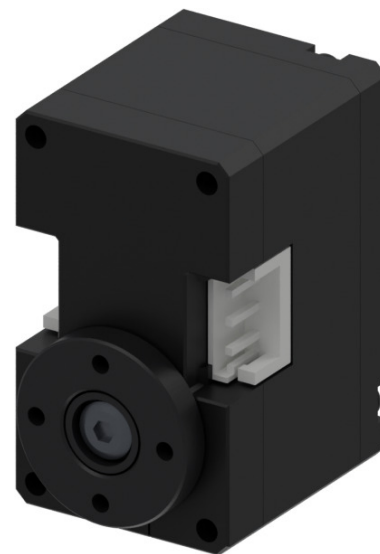
M2



M3



M4

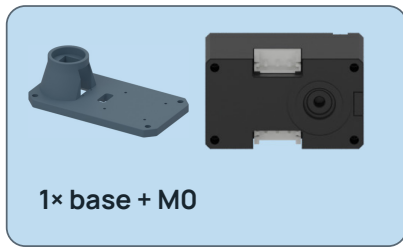


M5

Put these five assemblies in a tray. Later steps use one labelled motor at a time.

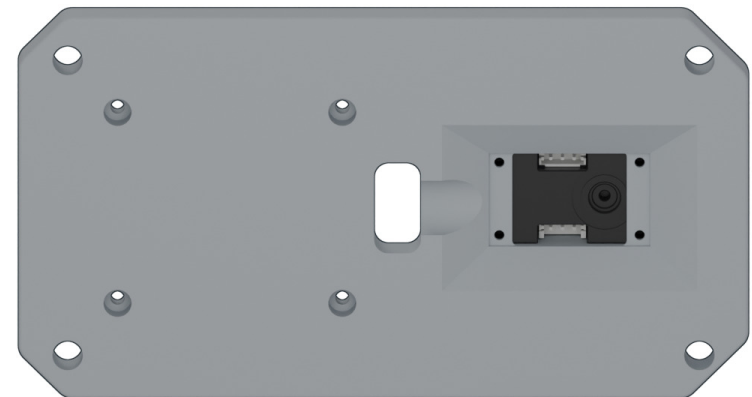
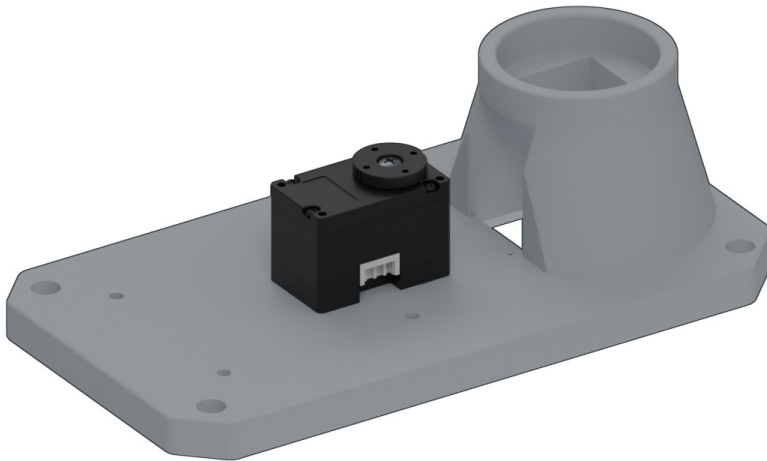
The tray count decreases as M1, M2, M3, M4 and M5 are installed.

M0 and M6 are separate: neither gets a rear wheel.



4

UNDERSIDE – turn the base over



Slide M0 horizontally through the open side of the base.

Fasten the four body holes from below. Wiring follows the mechanical assembly.

Next: insert the screws shown in the close-ups.

Prepared motors still in tray: M1 M2 M3 M4 M5



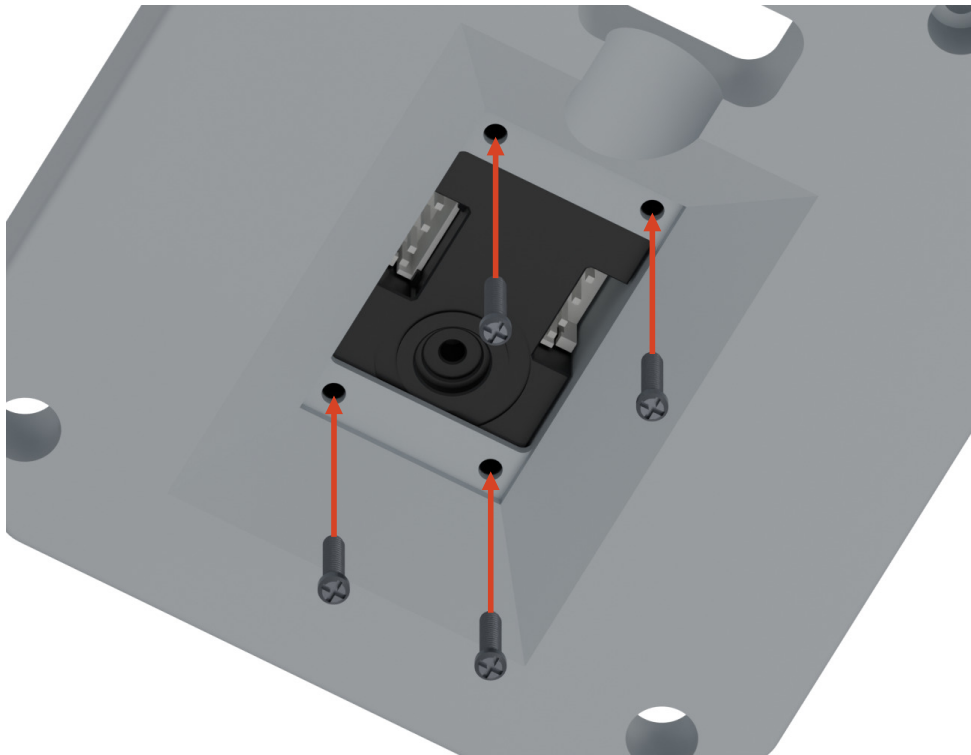
4× M2 × 8 TAP

Use this many on this face

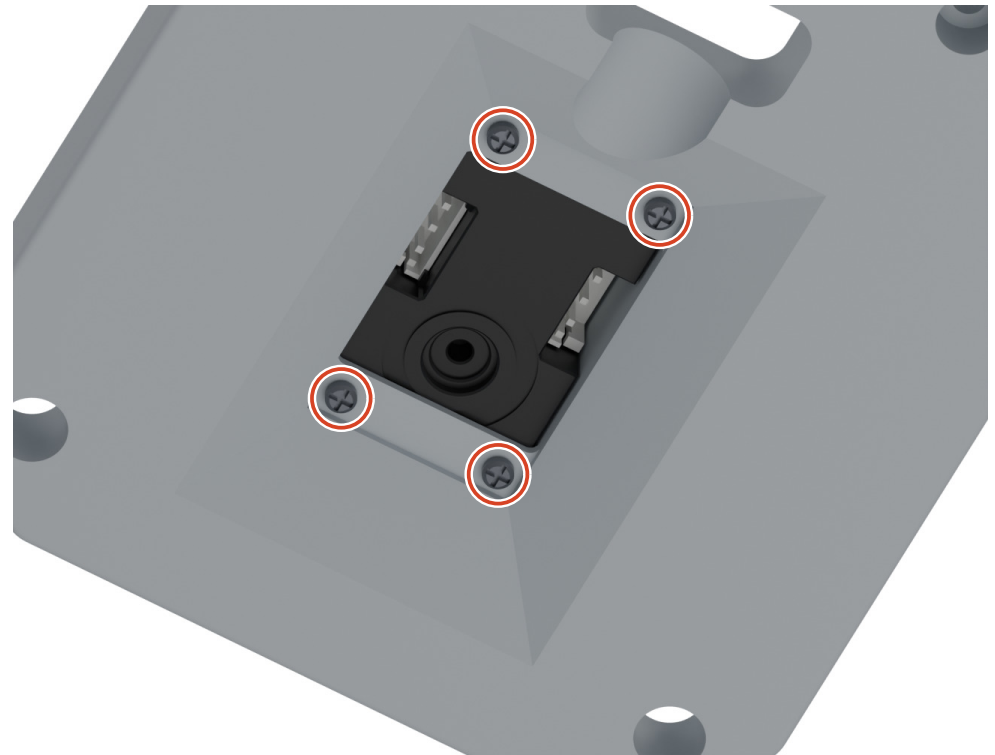
UNDERSIDE

5

INSERT – angled view



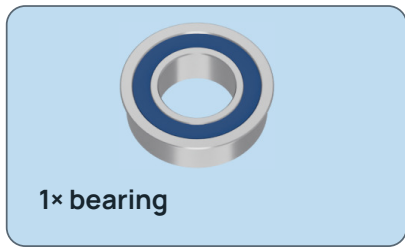
SEATED – screws fitted



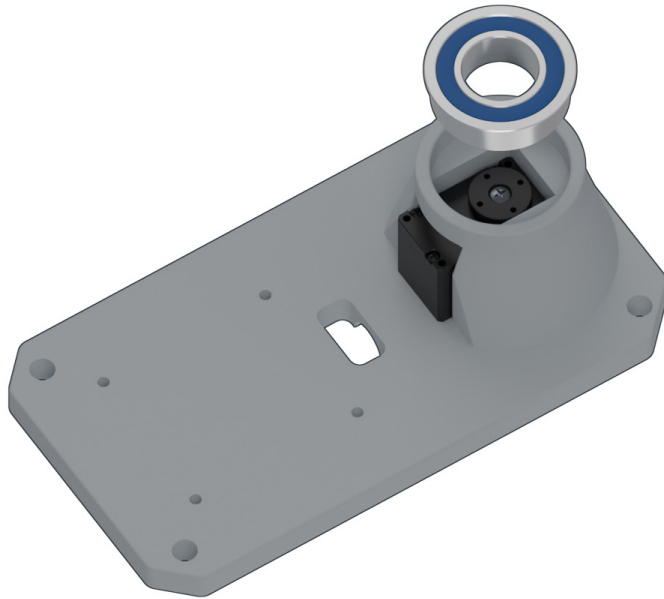
Insert 4 screws to fasten motor M0 to its printed holder.

Right: fit only the circled screws in this step. Other screws were fitted earlier.

Prepared motors still in tray: M1 M2 M3 M4 M5



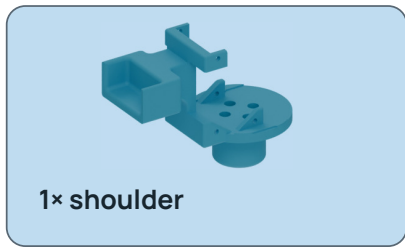
6



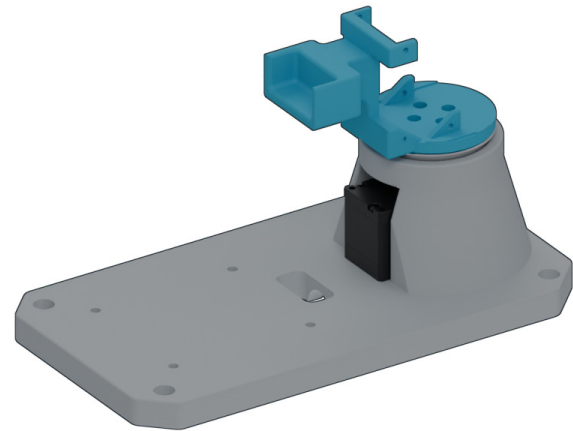
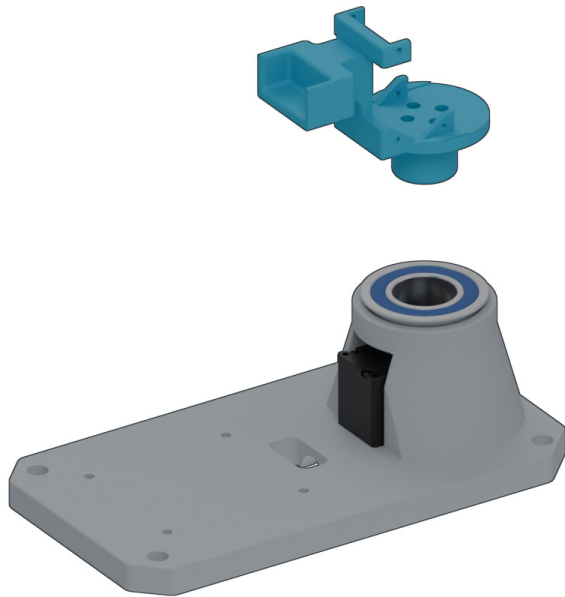
Seat the bearing fully on the circular ledge.

Fit it before the shoulder. Seat the bearing evenly.

Prepared motors still in tray: M1 M2 M3 M4 M5



7



Fit the empty shoulder onto M0.

Complete the following four-screw fastening step BEFORE fitting M1.

Next: insert the screws shown in the close-ups.

Prepared motors still in tray: M1 M2 M3 M4 M5



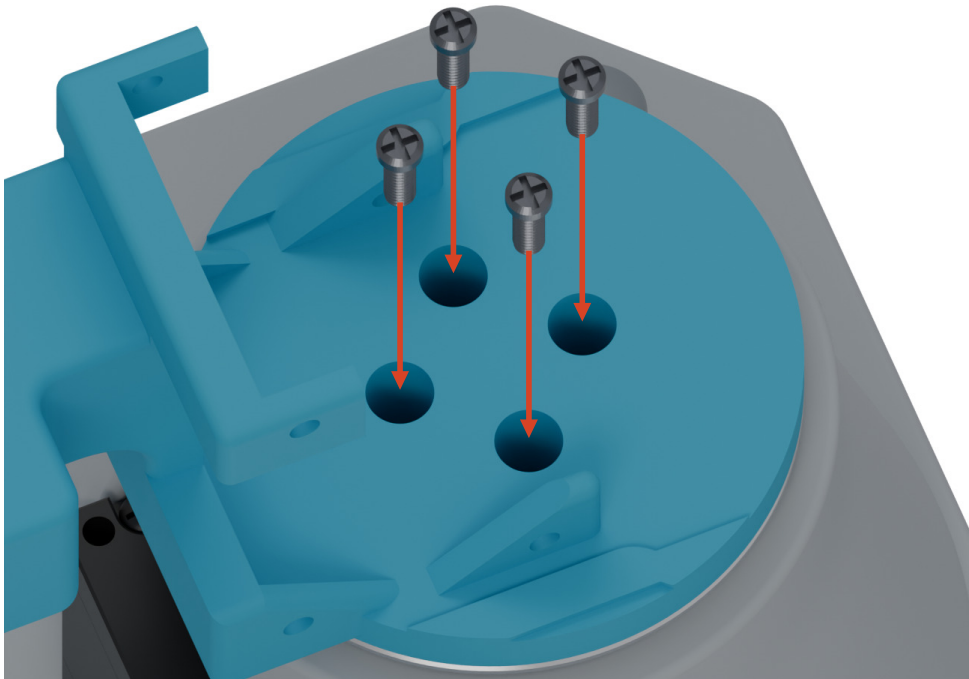
4× M2 × 6 TAP

Use this many on this face

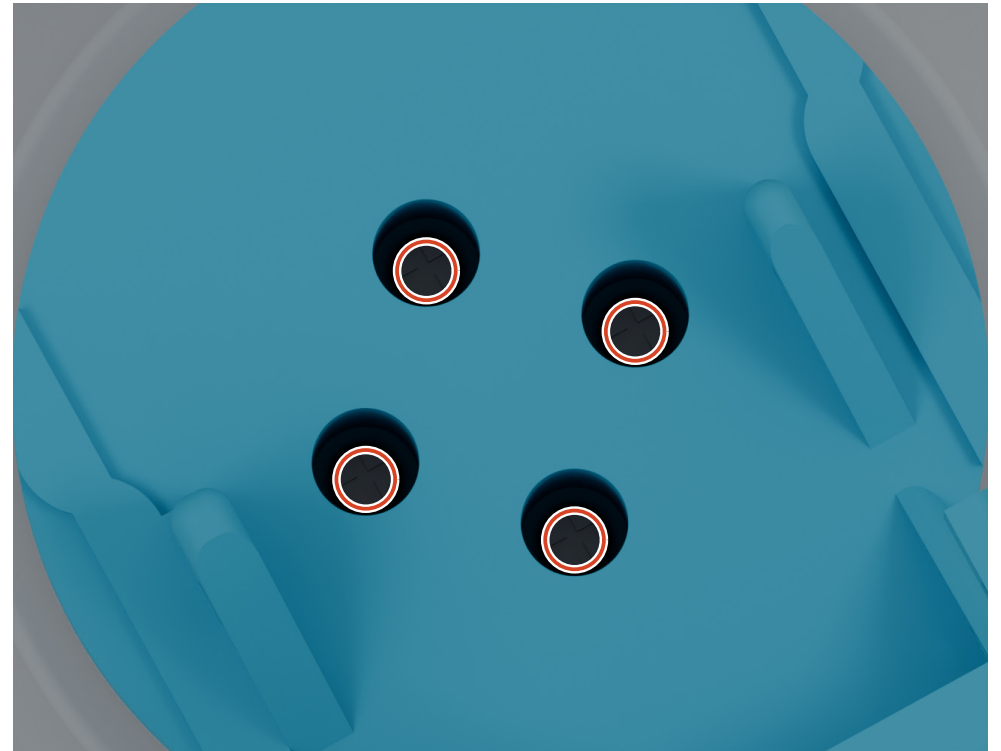
FRONT — horn side

8

INSERT — angled view



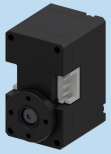
SEATED — screws fitted



Insert 4 screws to fasten the printed link to M0's front horn.

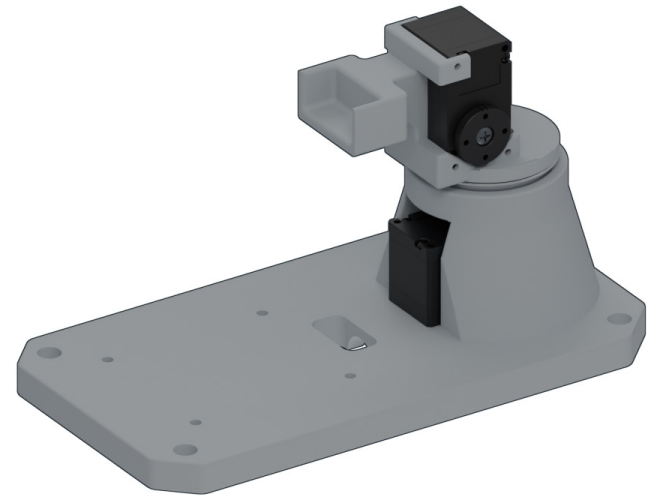
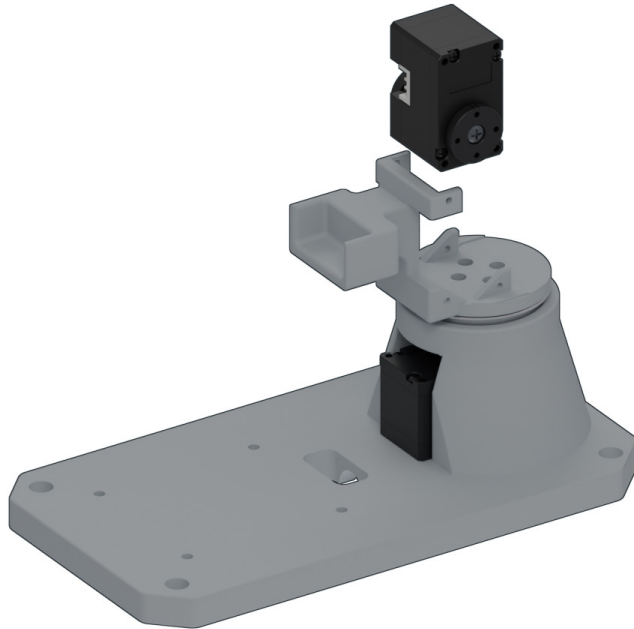
Right: all 4 heads sit inside the recesses. Keep previously installed screws fitted.

Prepared motors still in tray: M1 M2 M3 M4 M5



1× prepared M1

9



Lower prepared M1 through the open top of the shoulder holder.

Keep the rear wheel, cap and screw fitted; slide between the mounting tabs.

Next: insert the screws shown in the close-ups.

Prepared motors still in tray: M2 M3 M4 M5



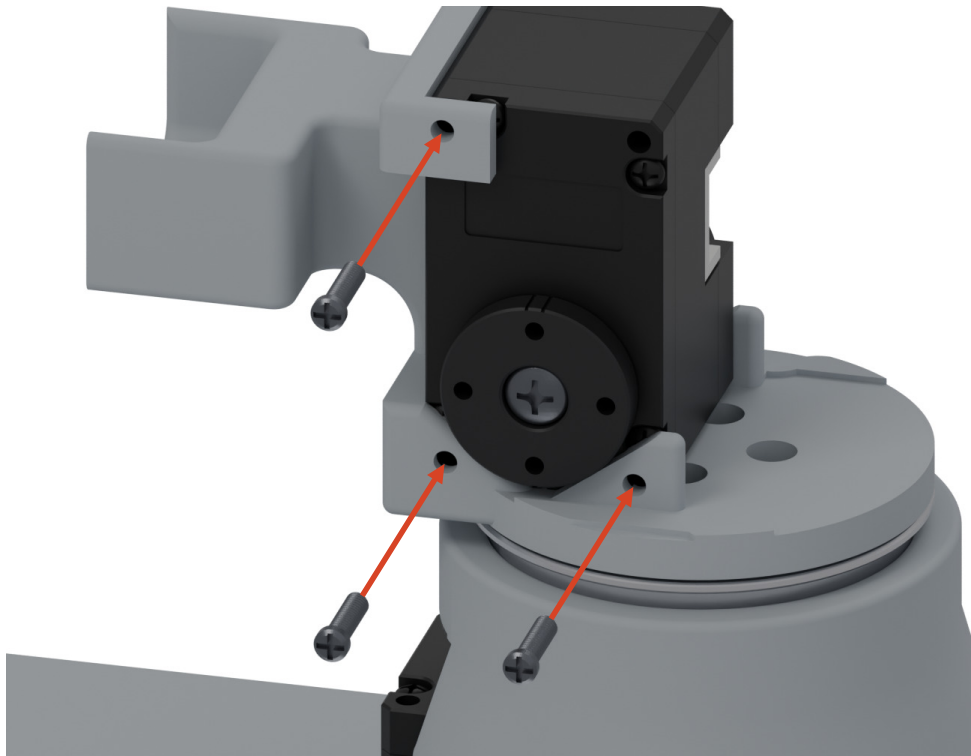
3× M2 × 8 TAP

Use this many on this face

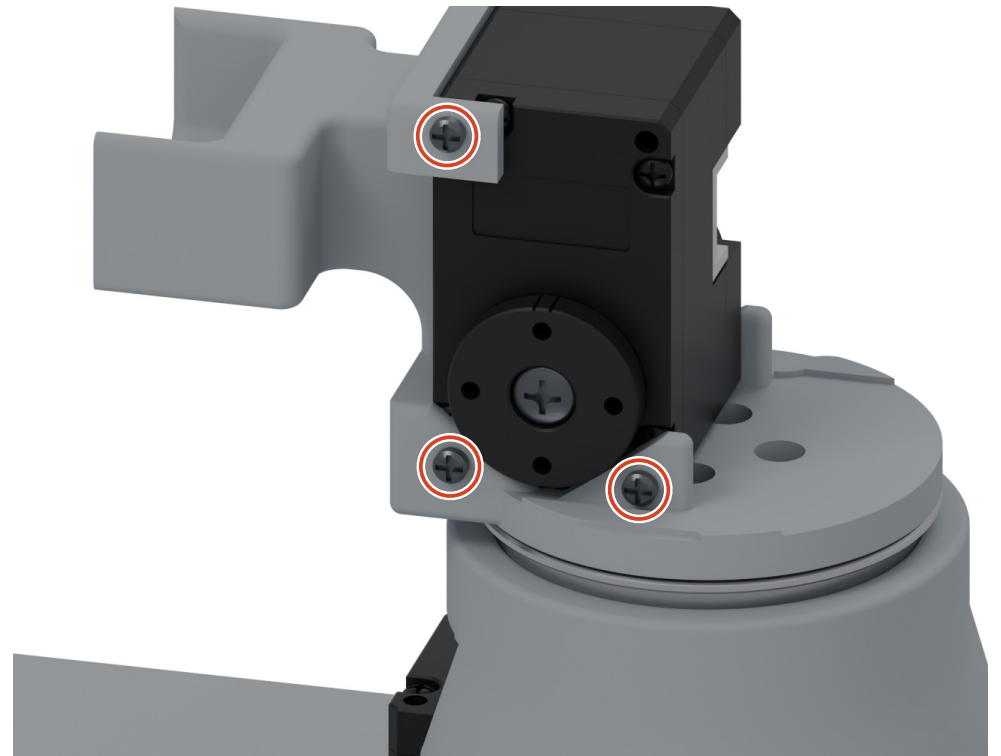
FRONT — horn side

10

INSERT — angled view



SEATED — screws fitted



Insert 3 screws to fasten motor M1 to its printed holder.

Right: fit only the circled screws in this step. Other screws were fitted earlier.

Prepared motors still in tray: M2 M3 M4 M5



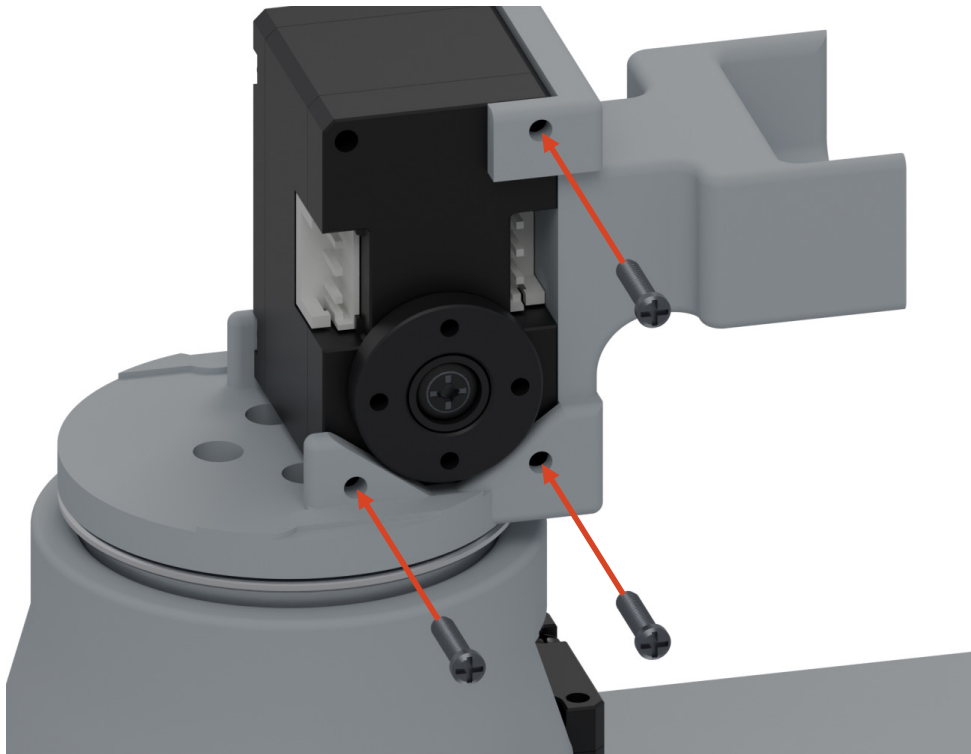
3× M2 × 8 TAP

Use this many on this face

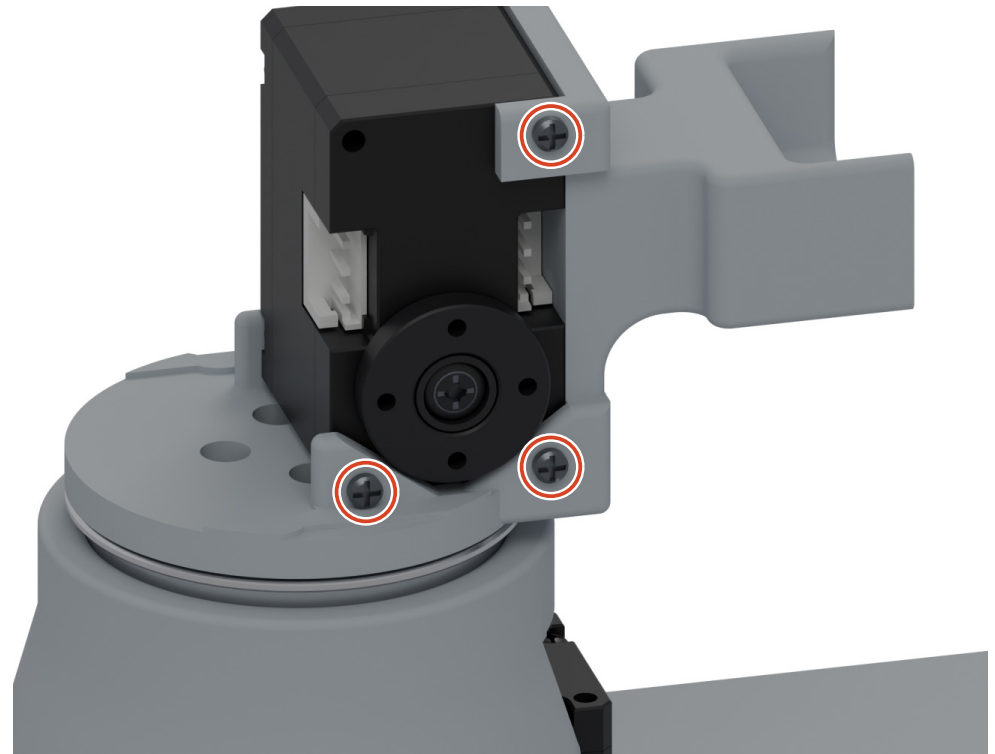
REAR – wheel side

11

INSERT – angled view



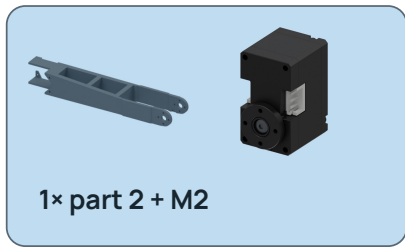
SEATED – screws fitted



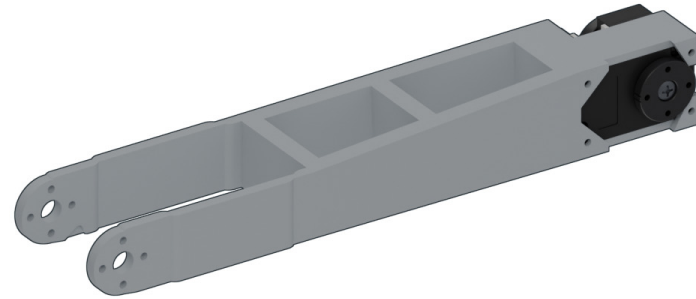
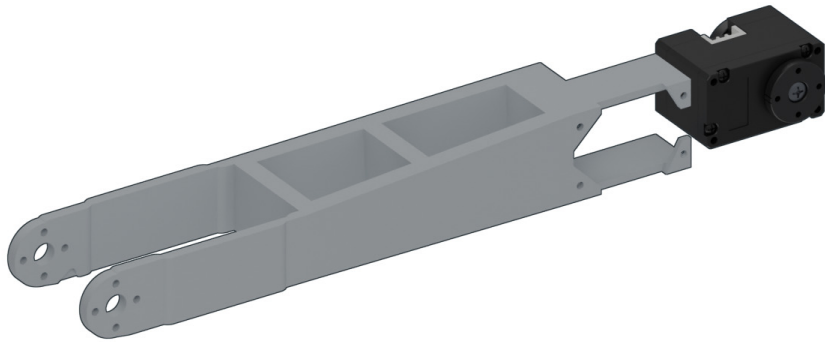
Insert 3 screws to fasten motor M1 to its printed holder.

Right: fit only the circled screws in this step. Other screws were fitted earlier.

Prepared motors still in tray: M2 M3 M4 M5



12



Slide prepared M2 into part 2 through its open end.

Keep the rear wheel, cap and screw fitted; do not push through the mounting faces.

Next: insert the screws shown in the close-ups.

Prepared motors still in tray: M3 M4 M5



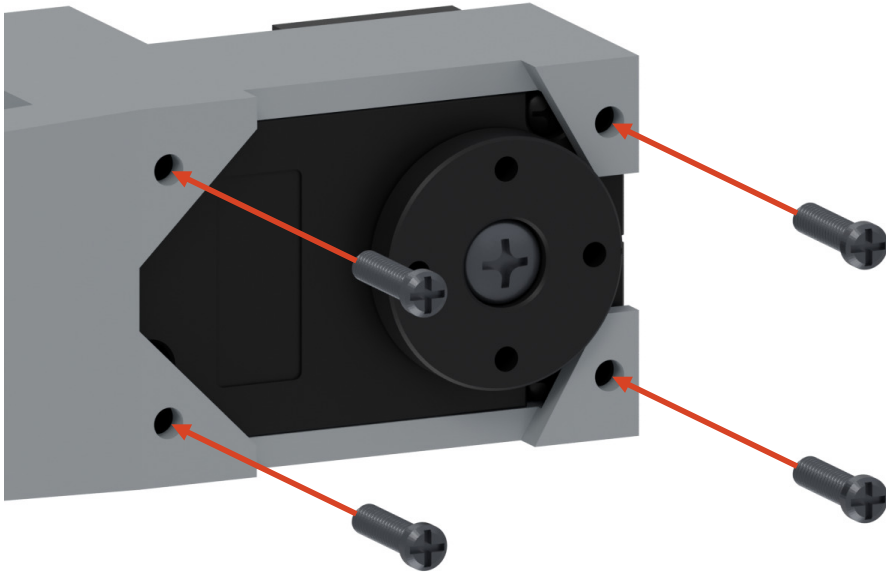
4× M2 × 8 TAP

Use this many on this face

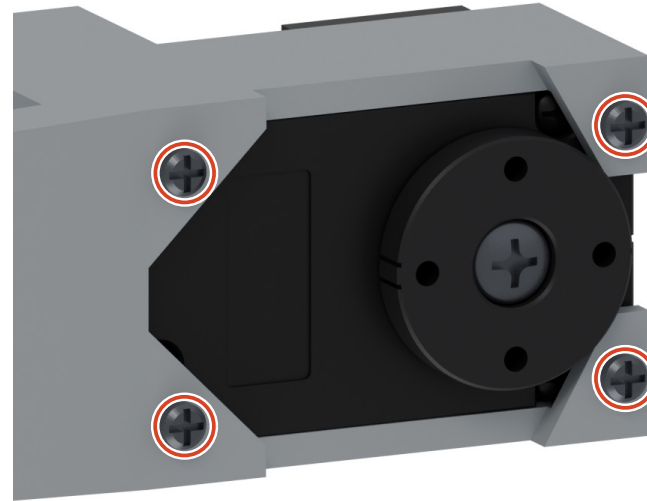
FRONT — horn side

13

INSERT — angled view



SEATED — screws fitted



Insert 4 screws to fasten motor M2 to its printed holder.

Right: fit only the circled screws in this step. Other screws were fitted earlier.

Prepared motors still in tray: M3 M4 M5



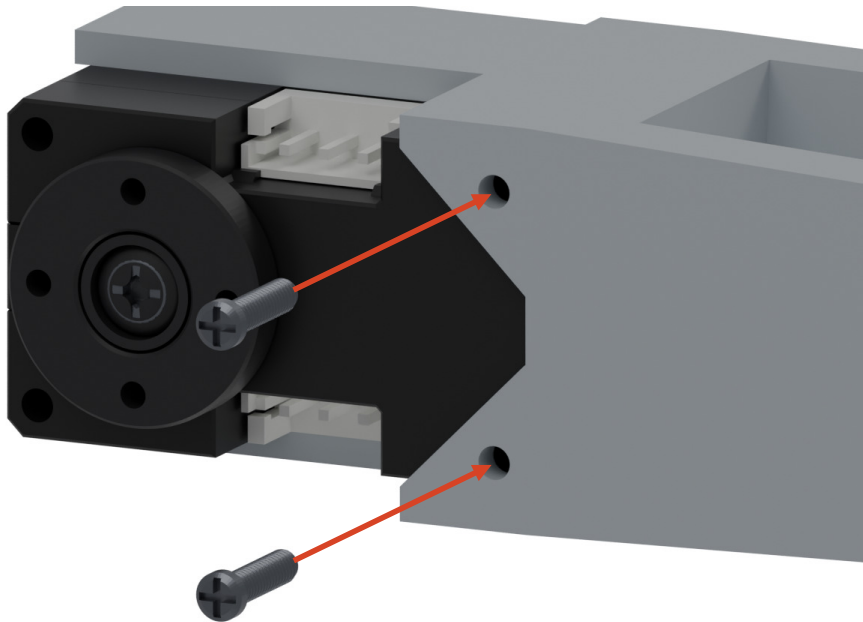
2× M2 × 8 TAP

Use this many on this face

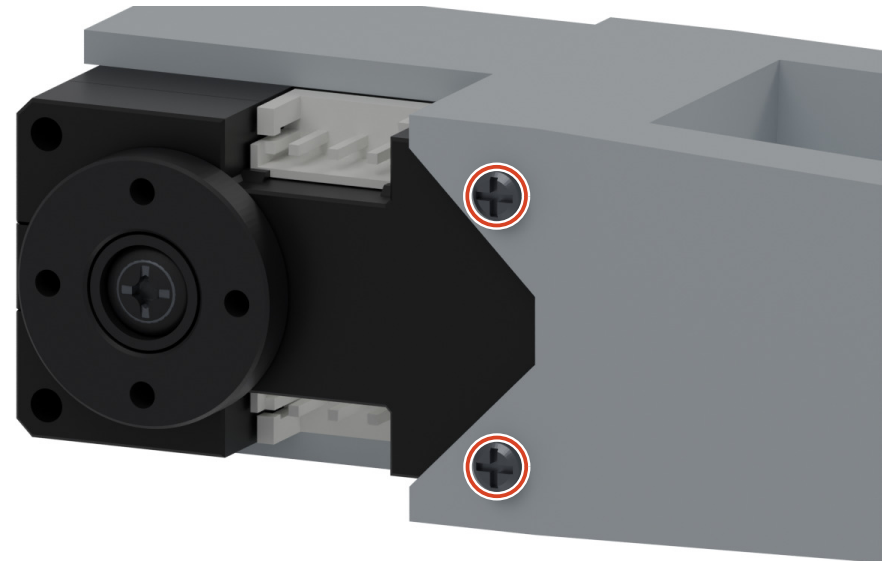
REAR – wheel side

14

INSERT – angled view



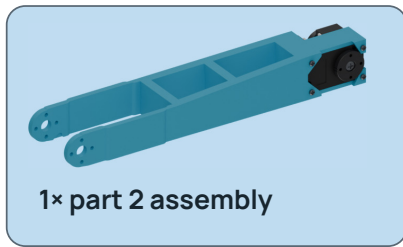
SEATED – screws fitted



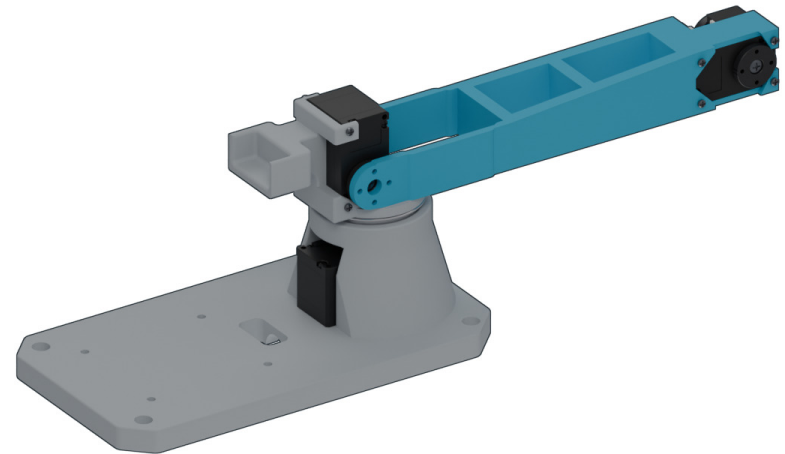
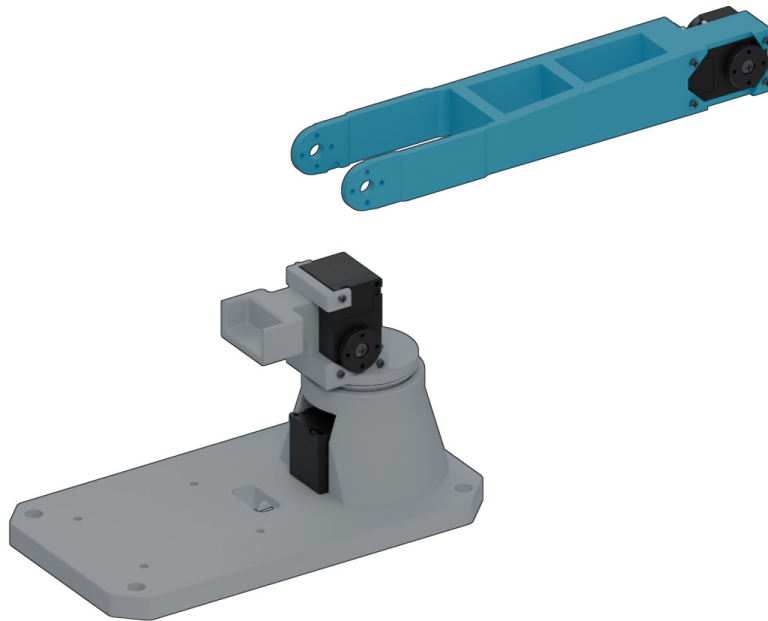
Insert 2 screws to fasten motor M2 to its printed holder.

Right: fit only the circled screws in this step. Other screws were fitted earlier.

Prepared motors still in tray: M3 M4 M5



15



Lower the part 2 assembly over M1's horn and rear wheel.

Fit two opposite screws per face; rotate the rear pair 90° from the front pair.

Next: insert the screws shown in the close-ups.

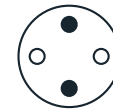
Prepared motors still in tray: M3 M4 M5



2× M2 × 6 TAP

Use this many on this face

FRONT — horn side



FRONT

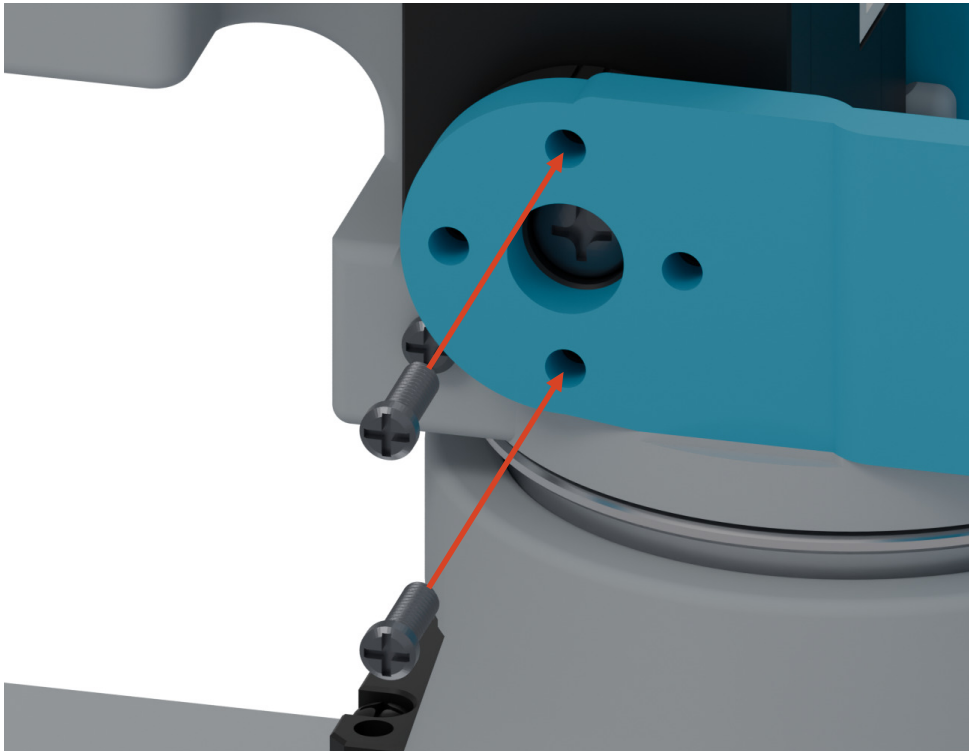


REAR

Opposite pairs · 90° apart

16

INSERT — angled view



SEATED — screws fitted



Insert 2 screws to fasten the printed link to M1's front horn.

Right: fit only the circled screws in this step. Other screws were fitted earlier.

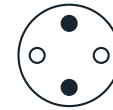
Prepared motors still in tray: M3 M4 M5



2× M2 × 6 TAP

Use this many on this face

REAR – wheel side



FRONT

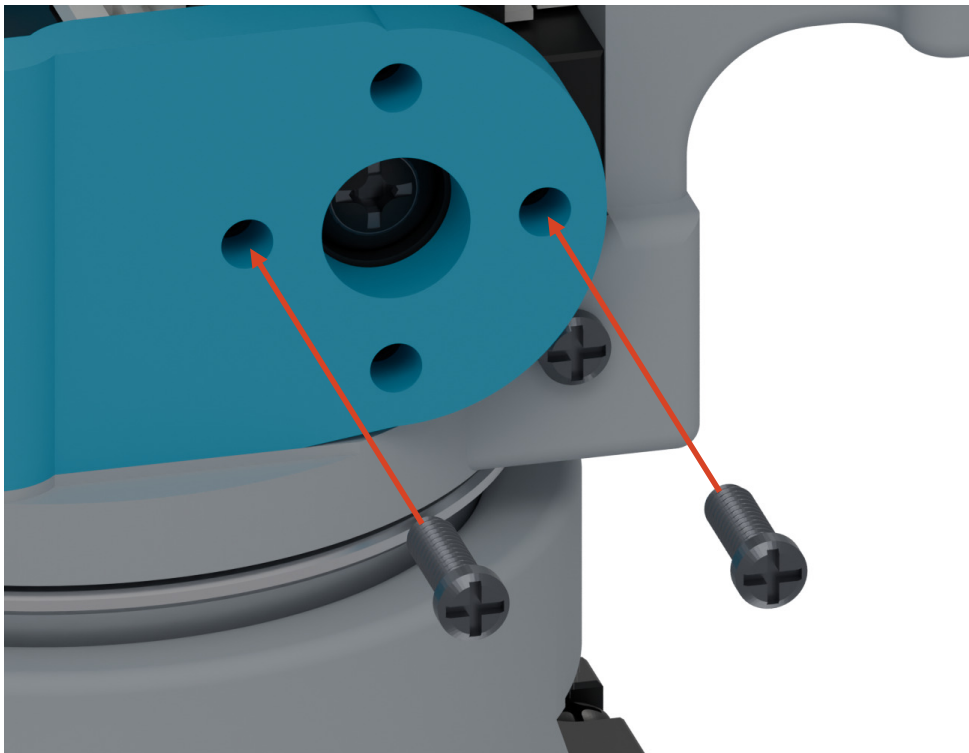


REAR

Opposite pairs · 90° apart

17

INSERT – angled view



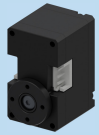
SEATED – screws fitted



Insert 2 screws to fasten the printed link to M1's rear wheel.

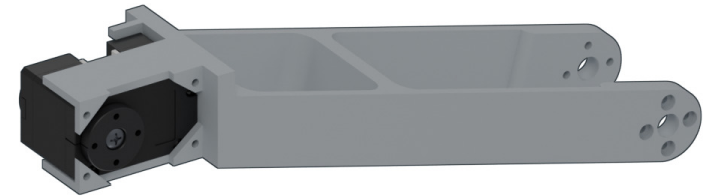
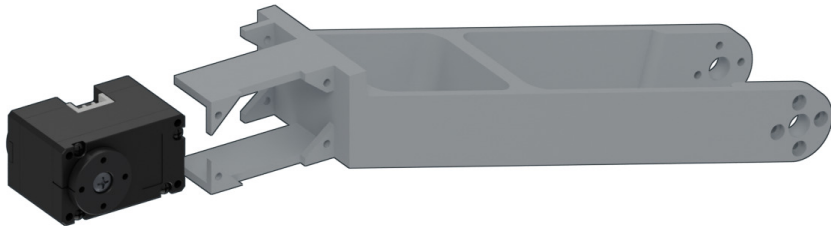
Right: fit only the circled screws in this step. Other screws were fitted earlier.

Prepared motors still in tray: M3 M4 M5



1× part 3 + M3

18



Slide prepared M3 into part 3 through its open end.

Keep the rear wheel, cap and screw fitted; do not push through the mounting faces.

Next: insert the screws shown in the close-ups.

Prepared motors still in tray: M4 M5



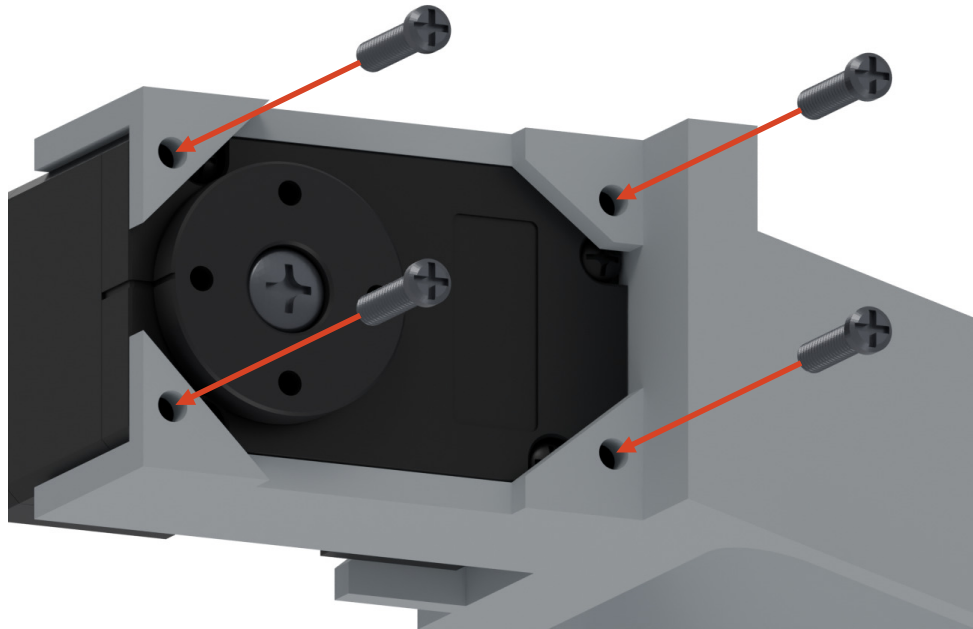
4× M2 × 8 TAP

Use this many on this face

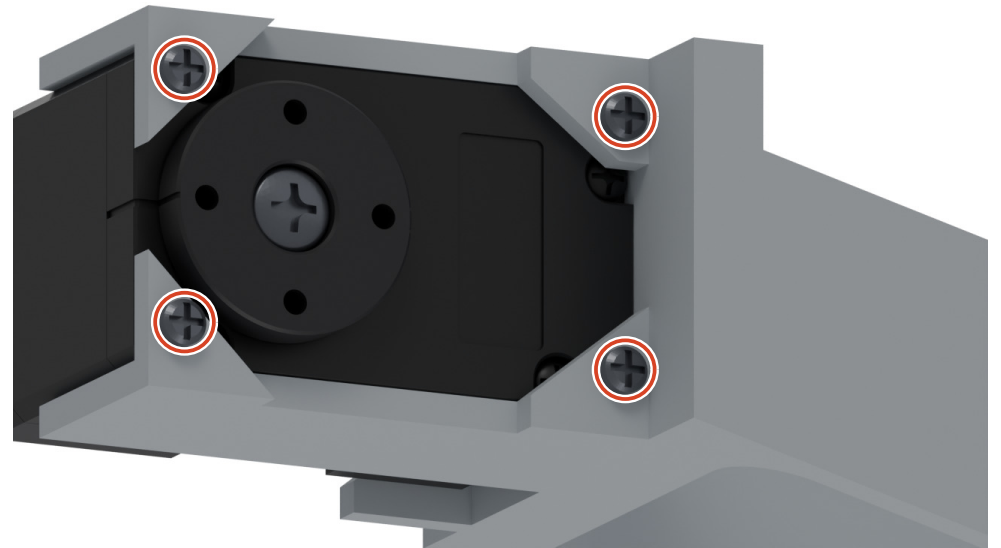
FRONT — horn side

19

INSERT — angled view



SEATED — screws fitted



Insert 4 screws to fasten motor M3 to its printed holder.

Right: fit only the circled screws in this step. Other screws were fitted earlier.

Prepared motors still in tray: M4 M5



2× M2 × 8 TAP

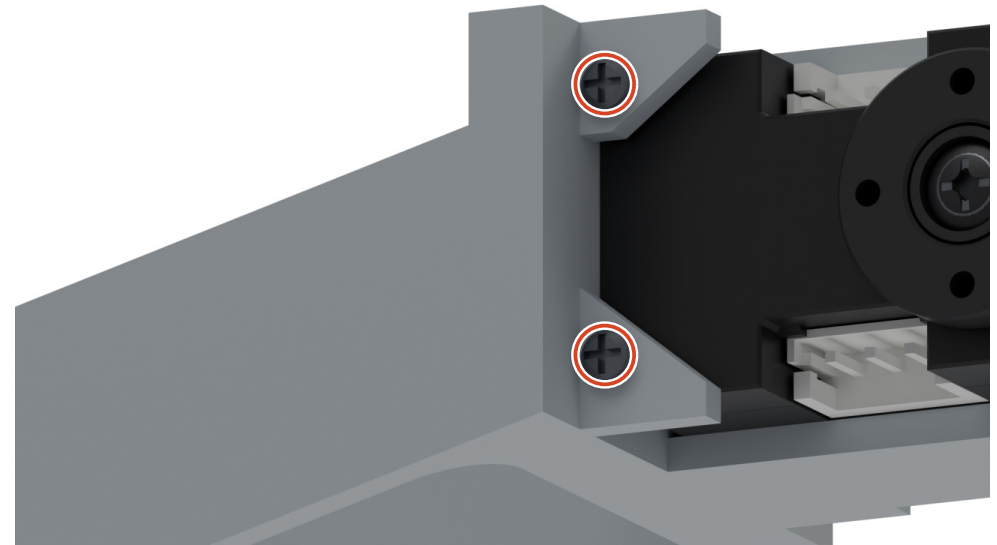
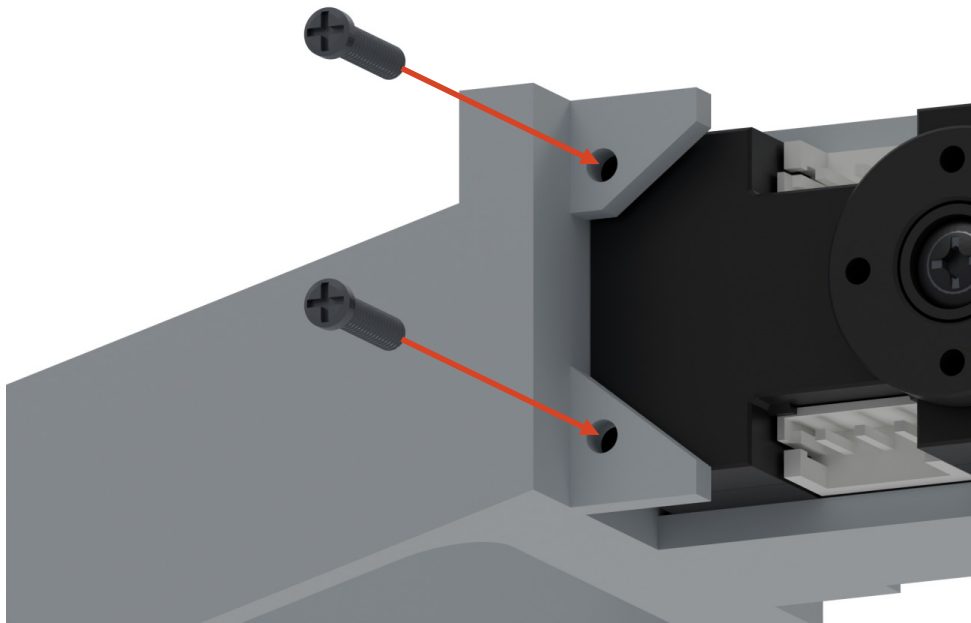
Use this many on this face

REAR – wheel side

20

INSERT – angled view

SEATED – screws fitted



Insert 2 screws to fasten motor M3 to its printed holder.

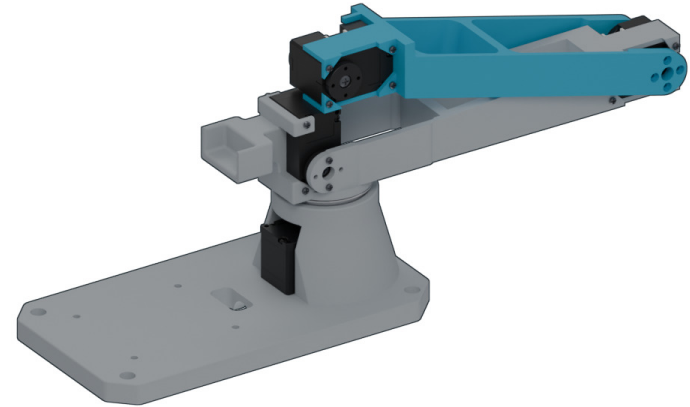
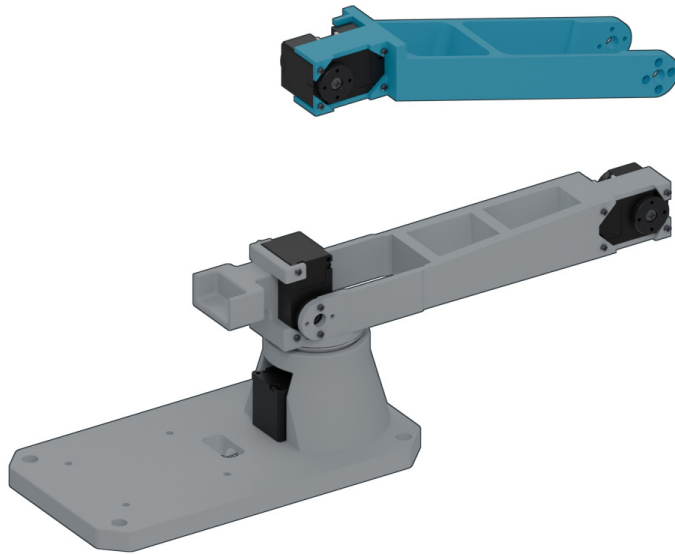
Right: fit only the circled screws in this step. Other screws were fitted earlier.

Prepared motors still in tray: M4 M5



1× part 3 assembly

21



Lower the part 3 assembly over M2's horn and rear wheel.

Fit two opposite screws per face; rotate the rear pair 90° from the front pair.

Next: insert the screws shown in the close-ups.

Prepared motors still in tray: M4 M5



2× M2 × 6 TAP

Use this many on this face

FRONT — horn side



FRONT

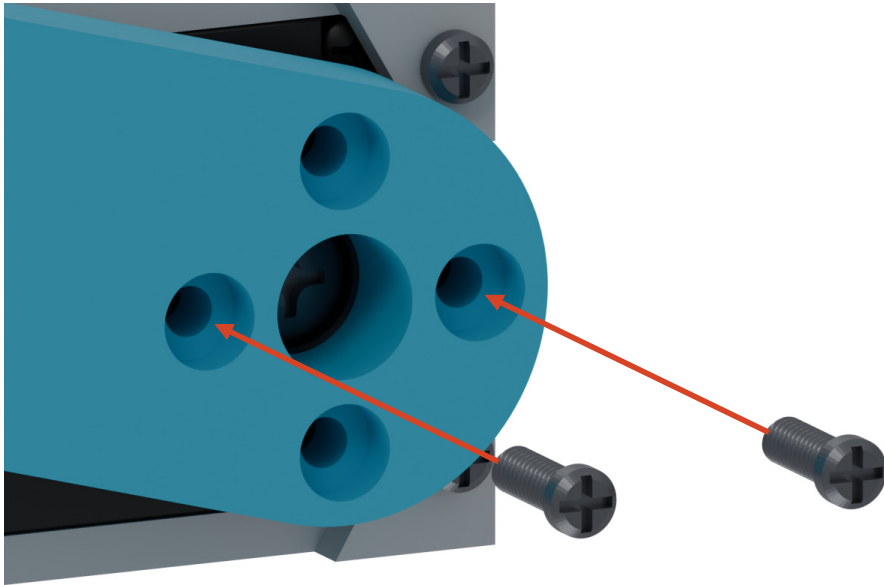


REAR

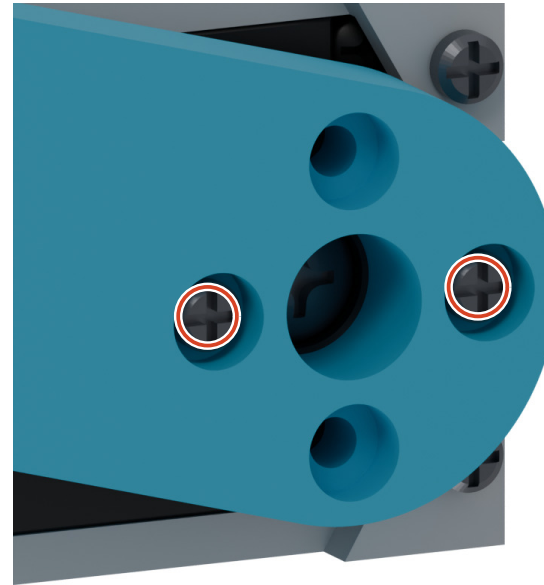
Opposite pairs · 90° apart

22

INSERT — angled view



SEATED — screws fitted



Insert 2 screws to fasten the printed link to M2's front horn.

Right: fit only the circled screws in this step. Other screws were fitted earlier.

Prepared motors still in tray: M4 M5



2× M2 × 6 TAP

Use this many on this face

REAR – wheel side



FRONT

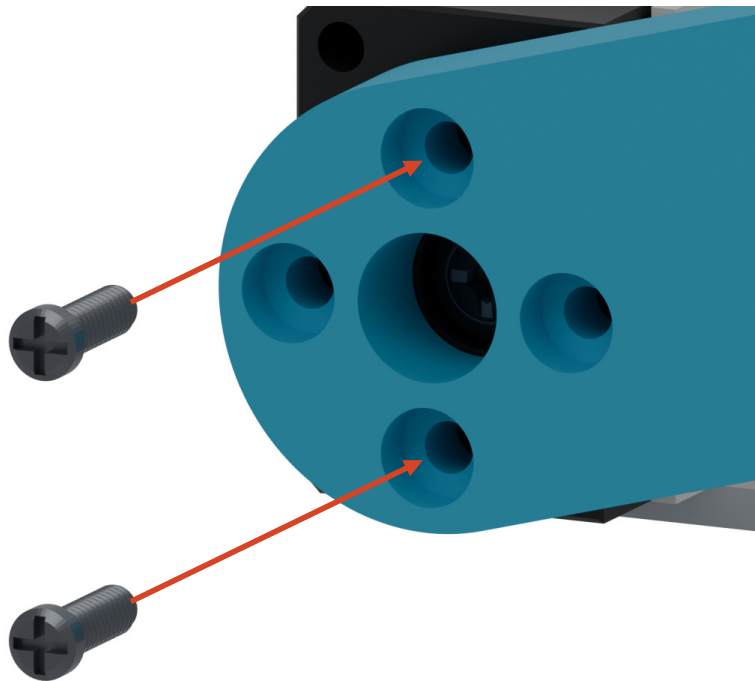


REAR

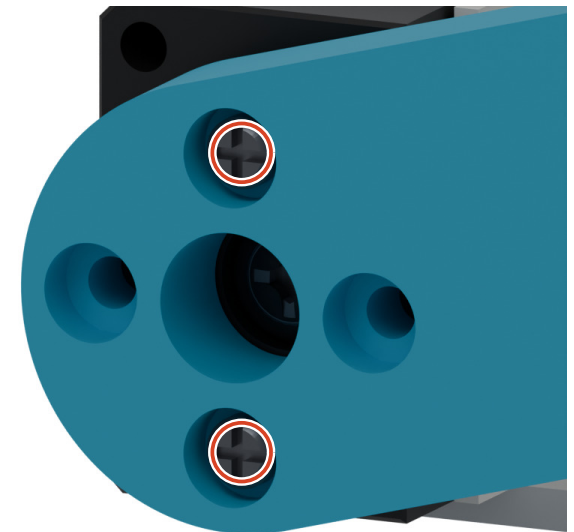
Opposite pairs · 90° apart

23

INSERT – angled view



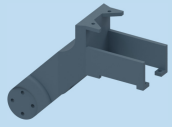
SEATED – screws fitted



Insert 2 screws to fasten the printed link to M2's rear wheel.

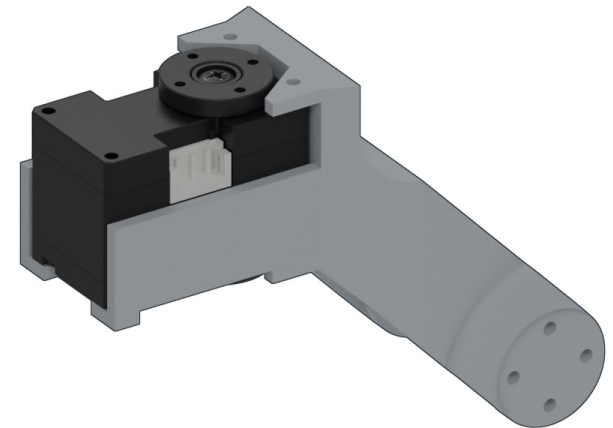
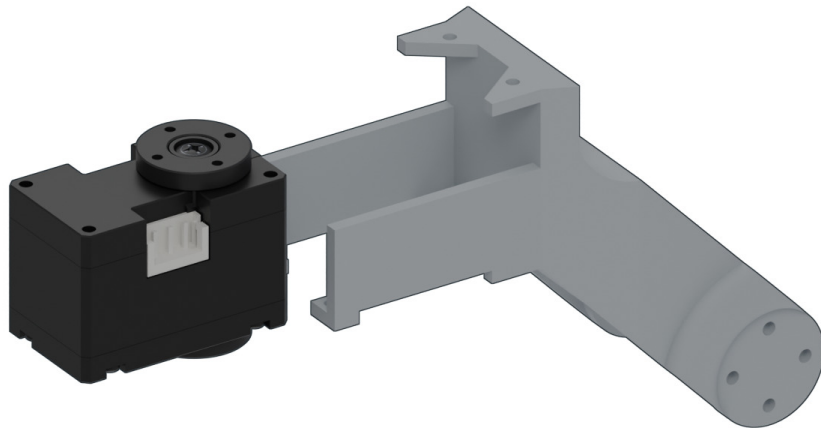
Right: fit only the circled screws in this step. Other screws were fitted earlier.

Prepared motors still in tray: M4 M5



1× part 4 + M4

24



Start a loose wrist: slide prepared M4 into part 4 from the left.

Use the open end between the tabs. Keep its rear wheel, cap and screw fitted.

Next: insert the screws shown in the close-ups.

Prepared motors still in tray: M5



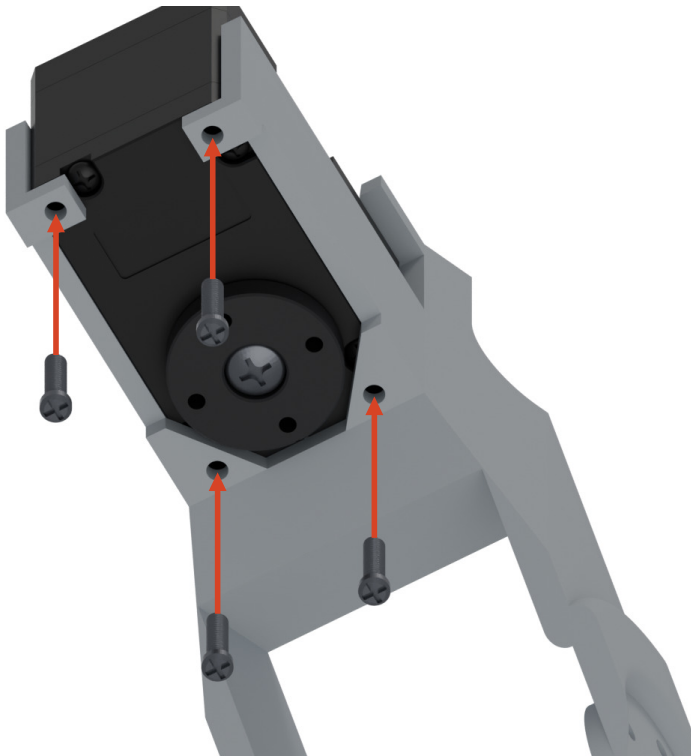
4× M2 × 8 TAP

Use this many on this face

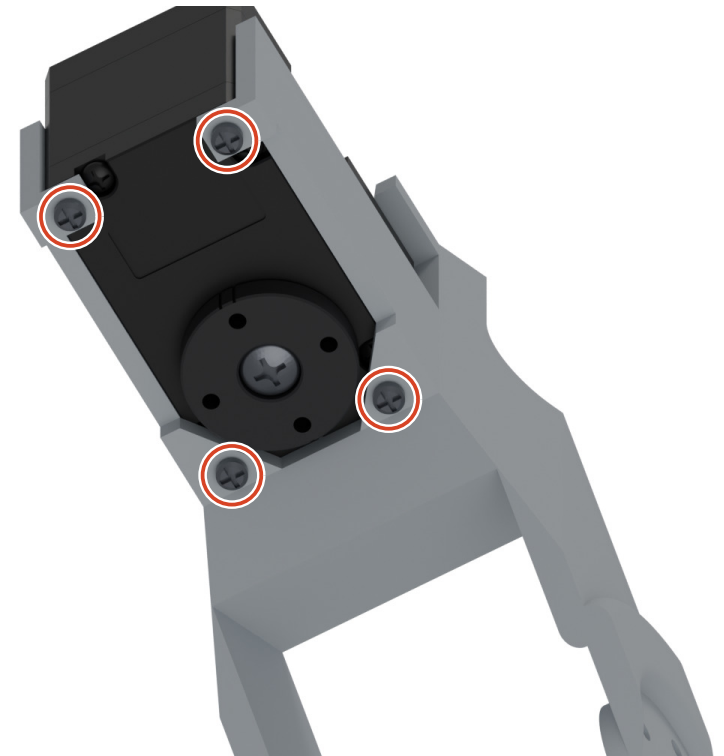
FRONT — horn side

25

INSERT — angled view



SEATED — screws fitted



Insert 4 screws to fasten motor M4 to its printed holder.

Right: fit only the circled screws in this step. Other screws were fitted earlier.

Prepared motors still in tray: M5



2× M2 × 8 TAP

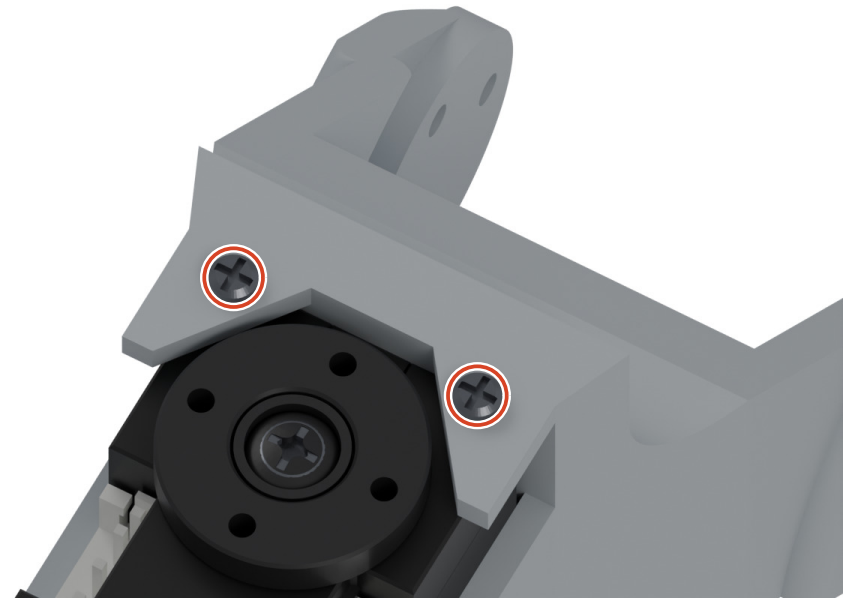
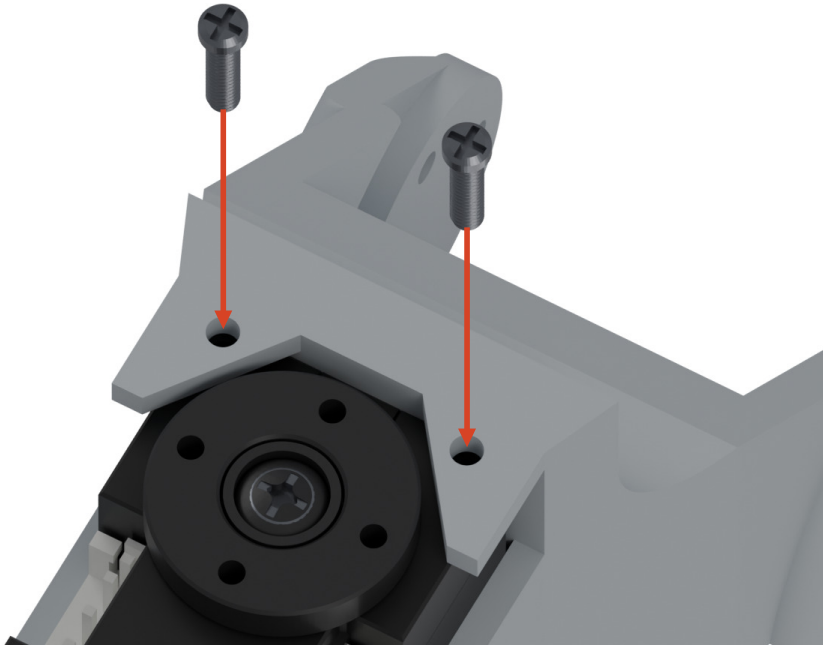
Use this many on this face

REAR – wheel side

26

INSERT – angled view

SEATED – screws fitted



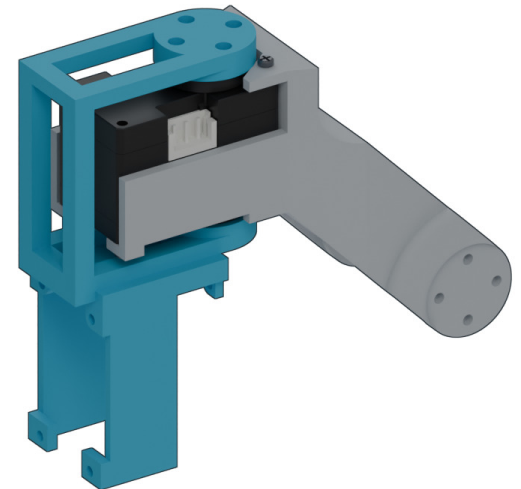
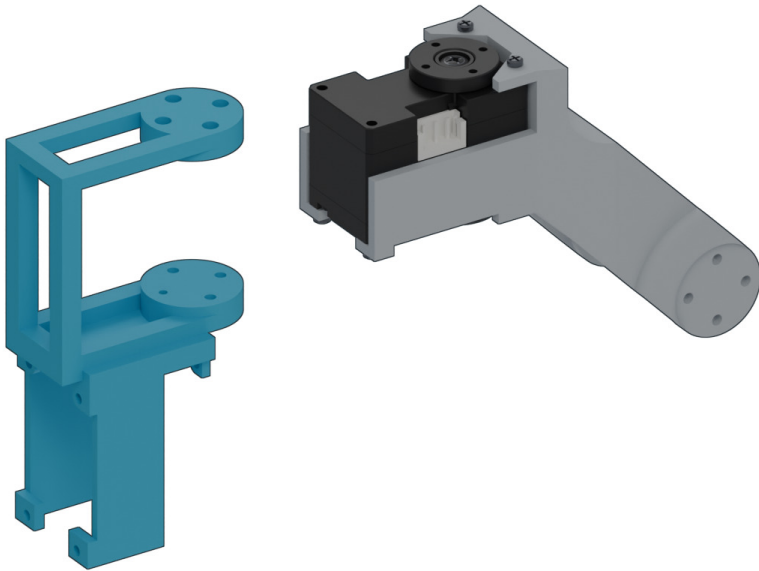
Insert 2 screws to fasten motor M4 to its printed holder.

Right: fit only the circled screws in this step. Other screws were fitted earlier.

Prepared motors still in tray: M5



27



Slide the empty wrist cage sideways onto M4 from the left.

Two opposite screws per face, rotated 90° between faces. Tighten before fitting M5.

Next: insert the screws shown in the close-ups.

Prepared motors still in tray: M5



2× M2 × 6 TAP

Use this many on this face

FRONT — horn side



FRONT

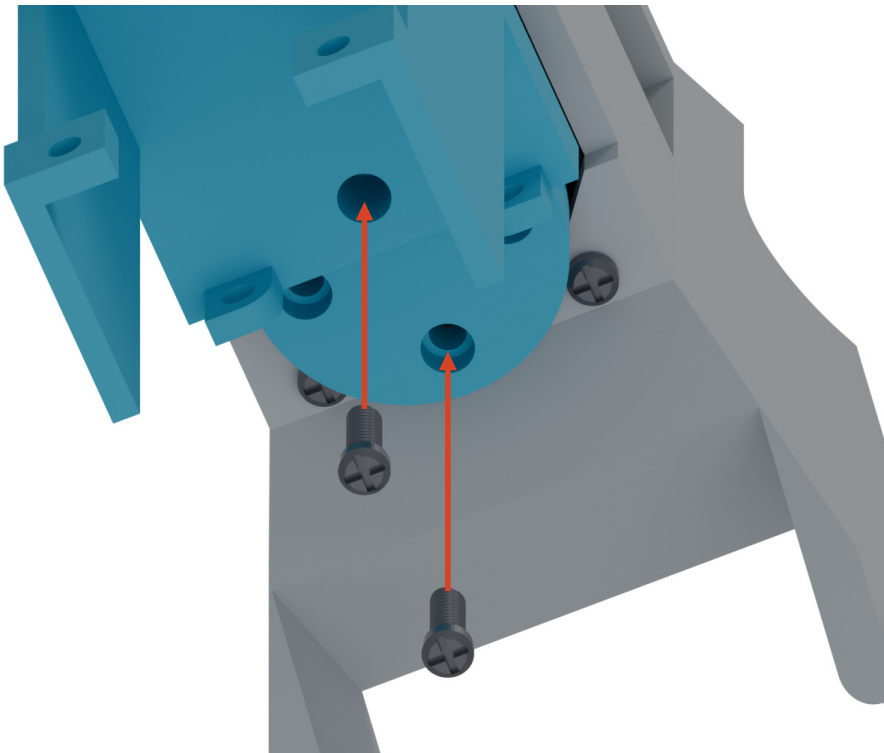


REAR

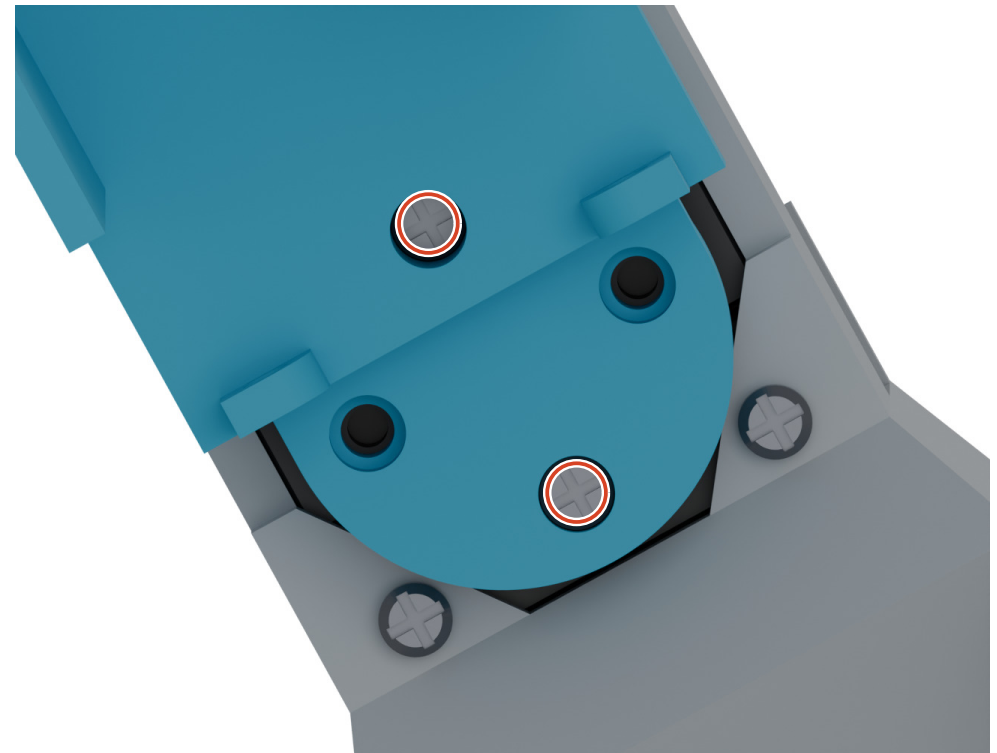
Opposite pairs · 90° apart

28

INSERT — angled view



SEATED — screws fitted



Insert 2 screws to fasten the printed link to M4's front horn.

Right: fit only the circled screws in this step. Other screws were fitted earlier.

Prepared motors still in tray: M5



2× M2 × 6 TAP

Use this many on this face

REAR – wheel side



FRONT

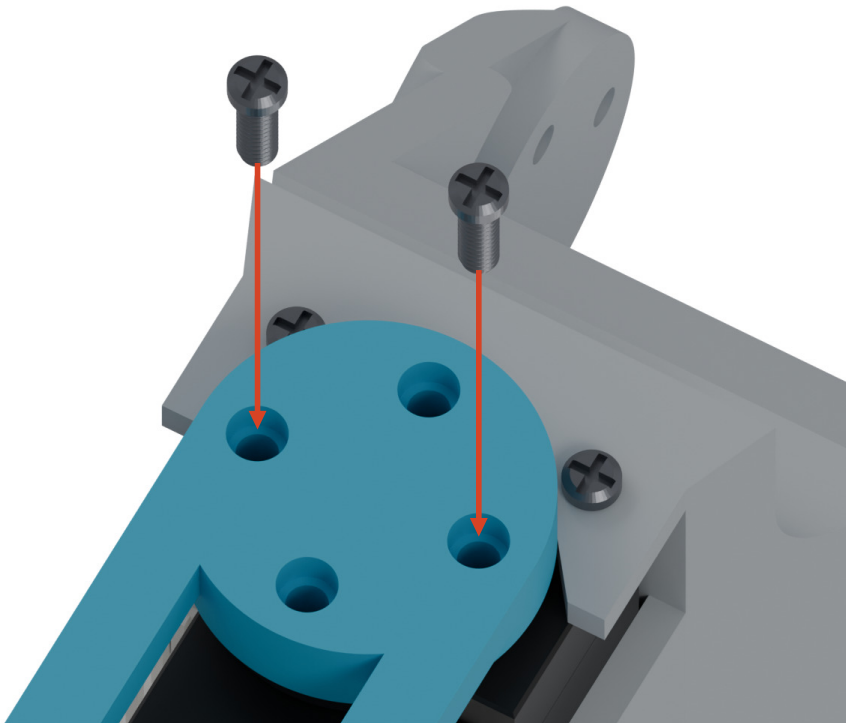


REAR

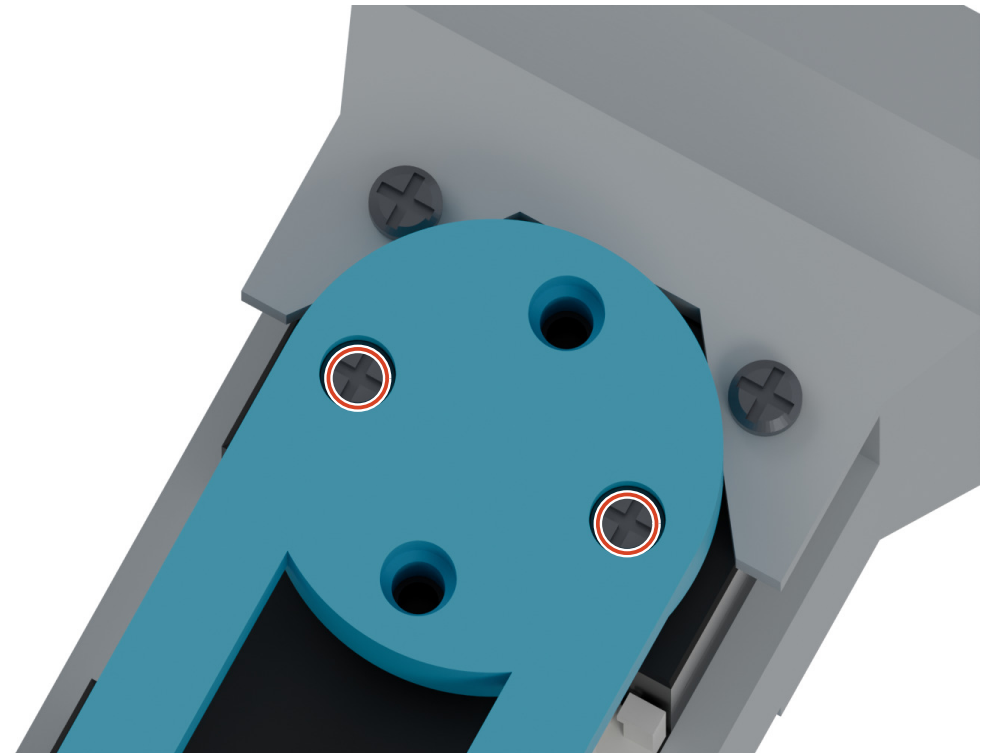
Opposite pairs · 90° apart

29

INSERT – angled view



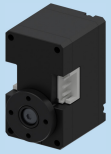
SEATED – screws fitted



Insert 2 screws to fasten the printed link to M4's rear wheel.

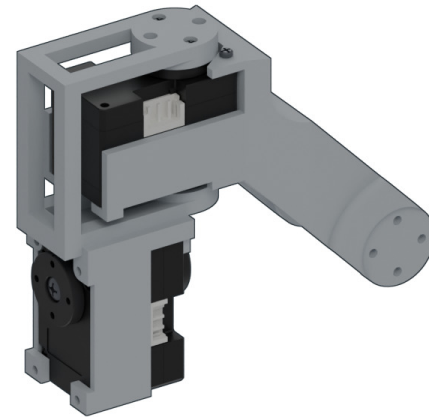
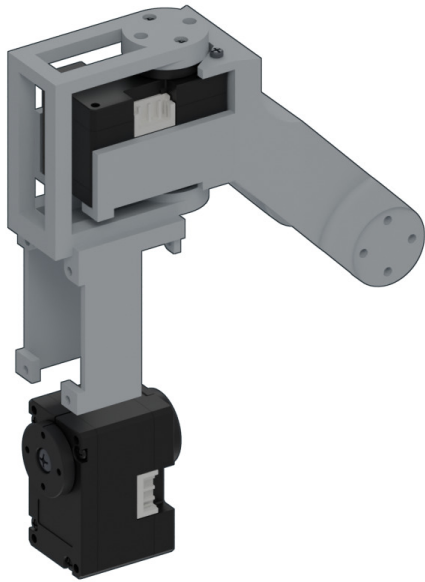
Right: fit only the circled screws in this step. Other screws were fitted earlier.

Prepared motors still in tray: M5



1× prepared M5

30



Slide prepared M5 upward into the open bottom of the cage.

Keep the wrist loose and retain M5's rear wheel, cap and screw.

Next: insert the screws shown in the close-ups.

All five prepared motors are now installed.



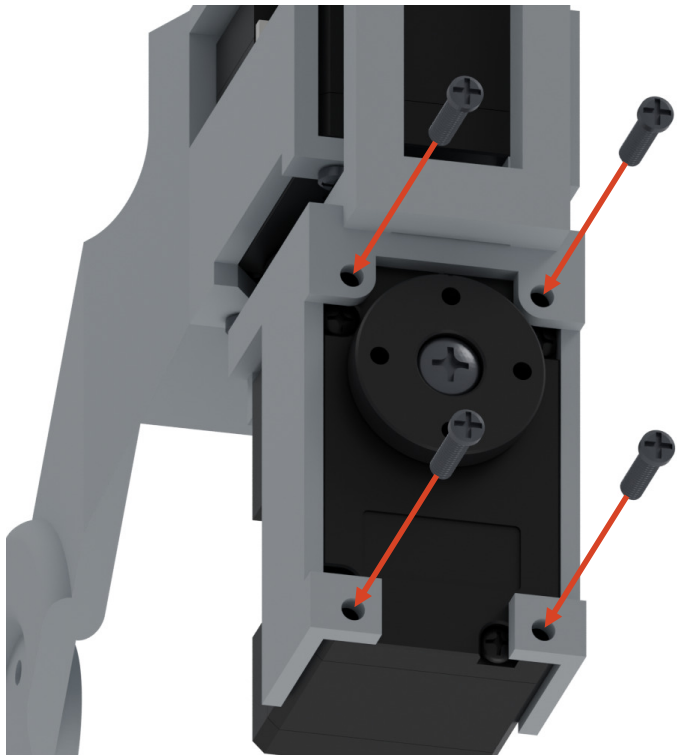
4× M2 × 8 TAP

Use this many on this face

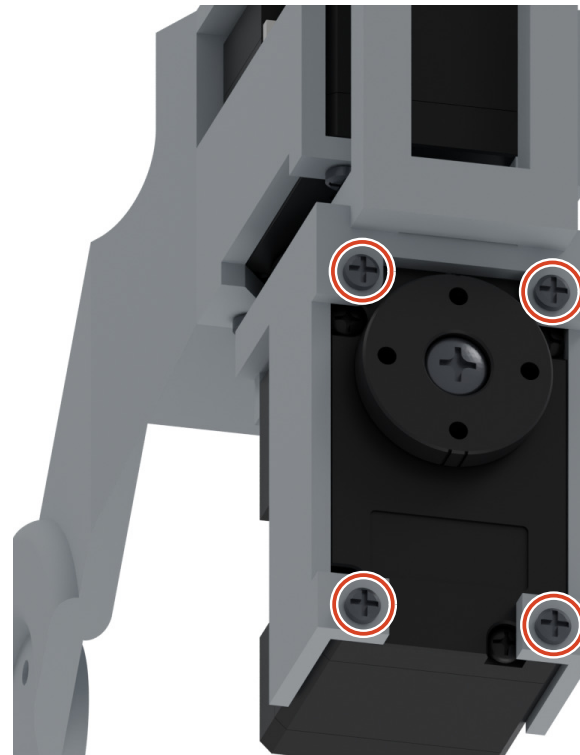
FRONT — horn side

31

INSERT — angled view



SEATED — screws fitted



Insert 4 screws to fasten motor M5 to its printed holder.

Right: fit only the circled screws in this step. Other screws were fitted earlier.

All five prepared motors are now installed.



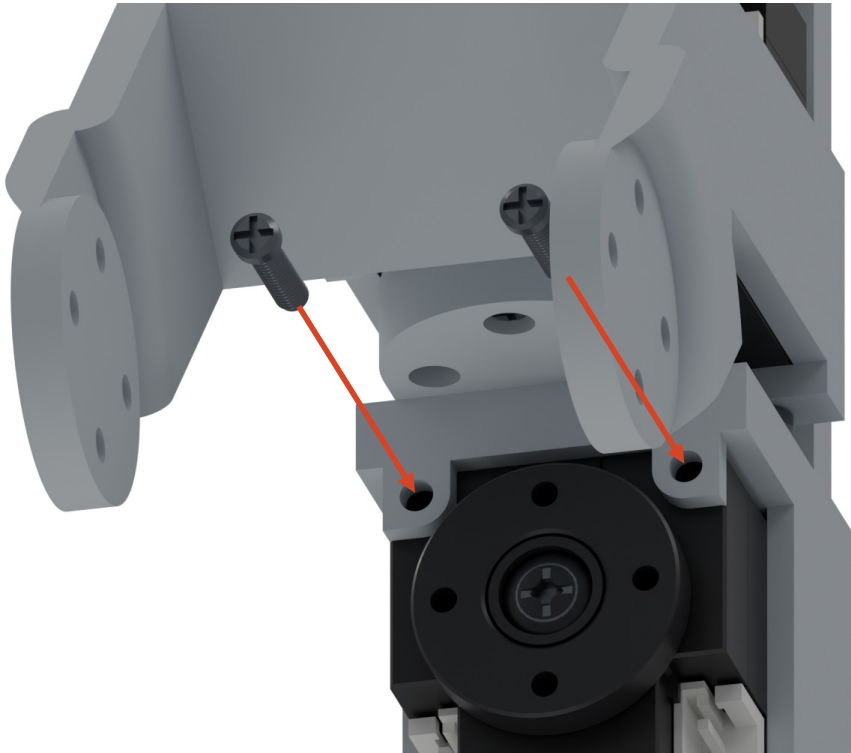
2× M2 × 8 TAP

Use this many on this face

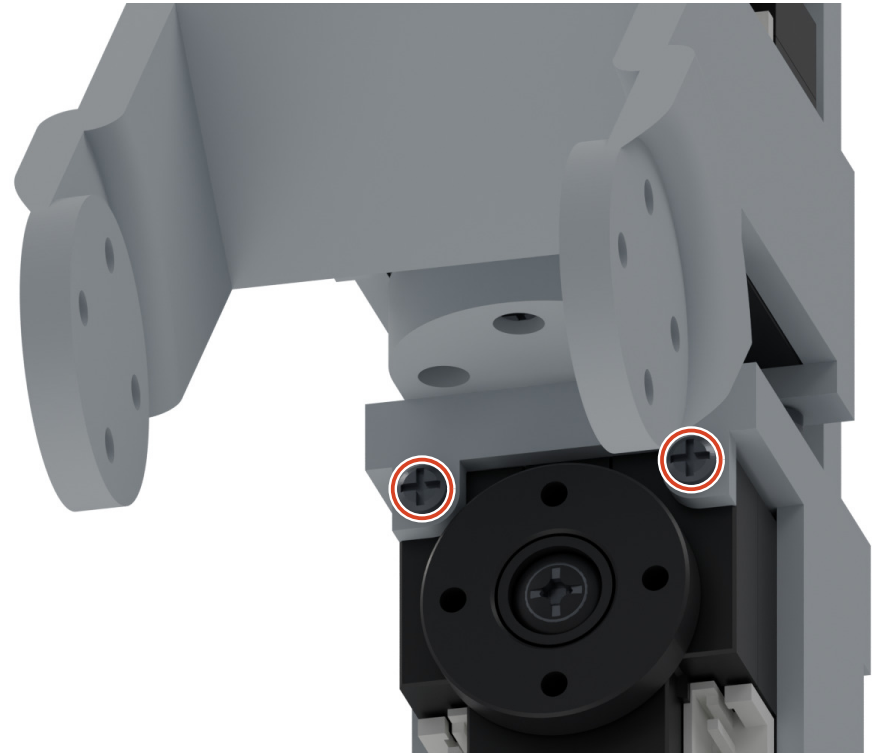
REAR – wheel side

32

INSERT – angled view



SEATED – screws fitted



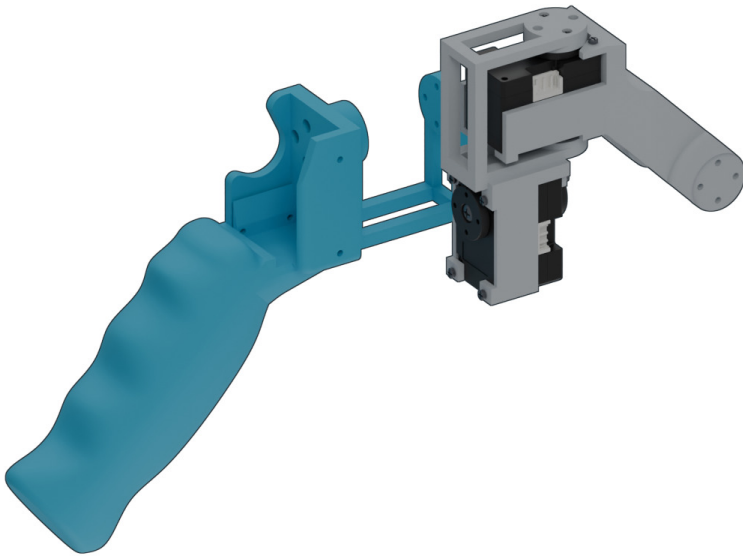
Insert 2 screws to fasten motor M5 to its printed holder.

Right: fit only the circled screws in this step. Other screws were fitted earlier.

All five prepared motors are now installed.



33



Slide the empty handle sideways over M5's horn and rear wheel.

Two opposite screws per face, rotated 90° between faces. Tighten before fitting M6.

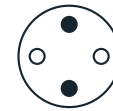
Next: insert the screws shown in the close-ups.



2× M2 × 6 TAP

Use this many on this face

FRONT — horn side



FRONT



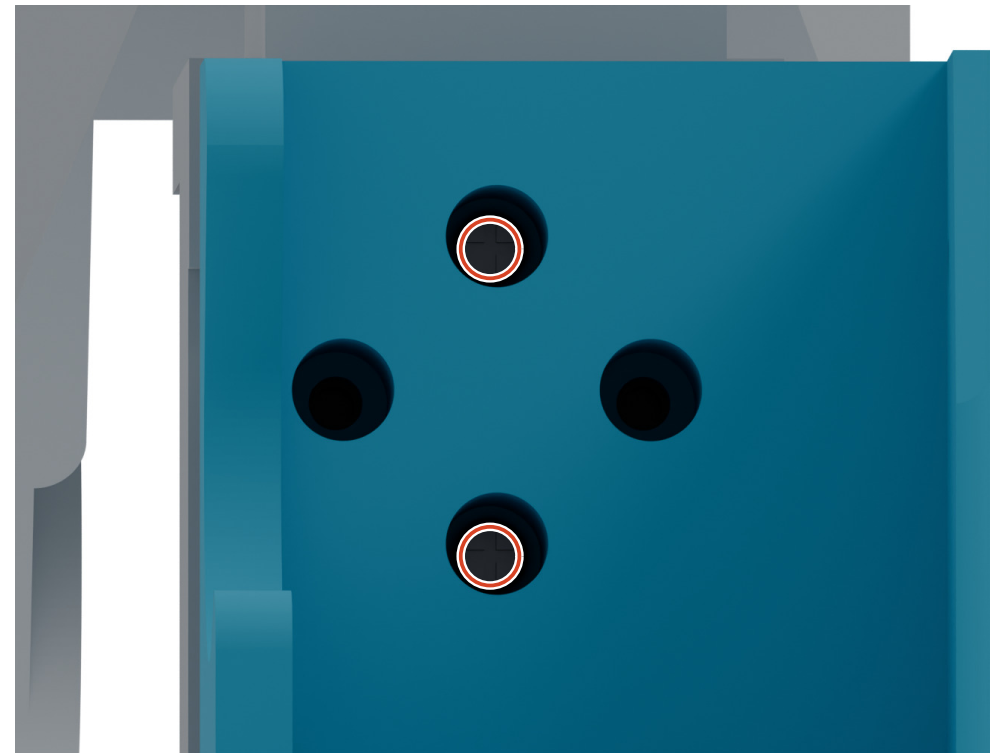
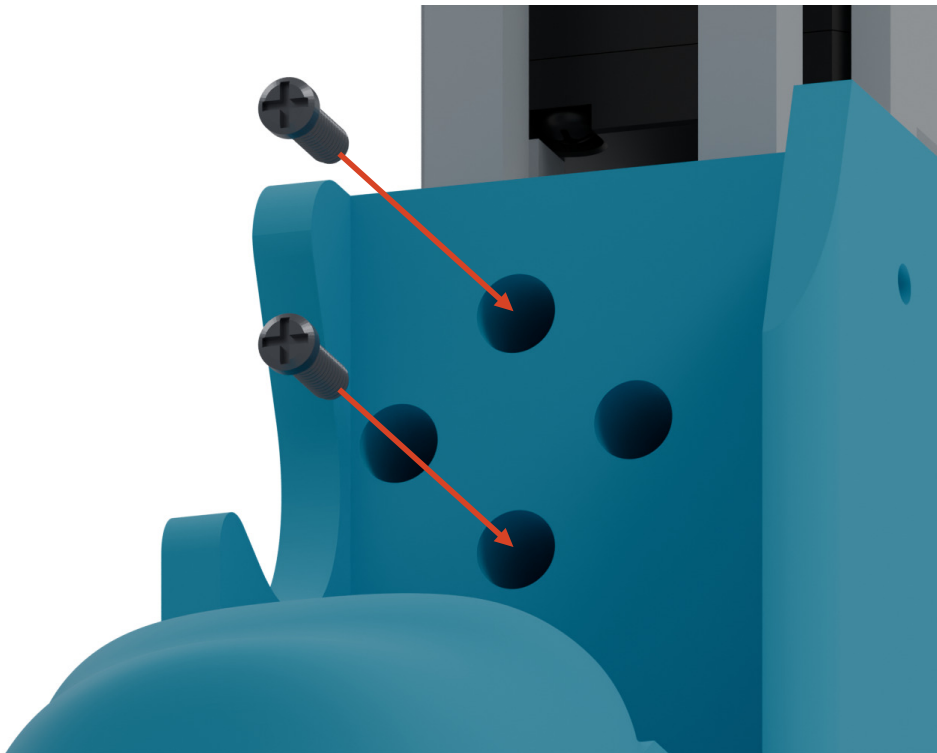
REAR

Opposite pairs · 90° apart

34

INSERT — angled view

SEATED — screws fitted



Insert 2 screws to fasten the printed link to M5's front horn.

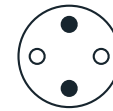
Right: all 2 heads sit inside the recesses. Keep previously installed screws fitted.



2× M2 × 6 TAP

Use this many on this face

REAR – wheel side



FRONT

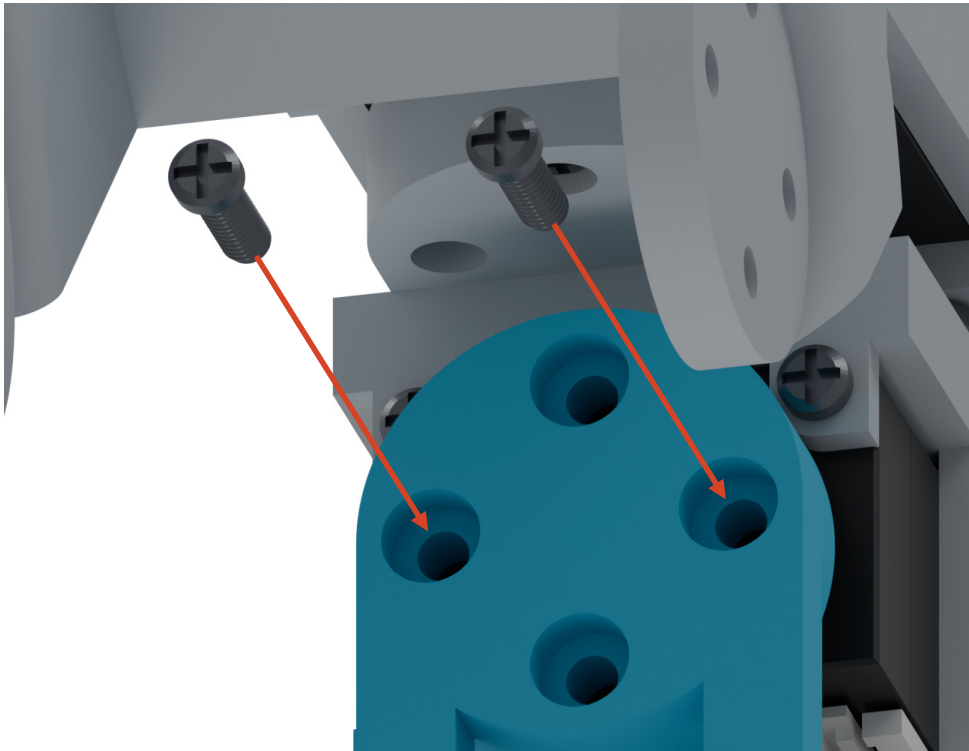


REAR

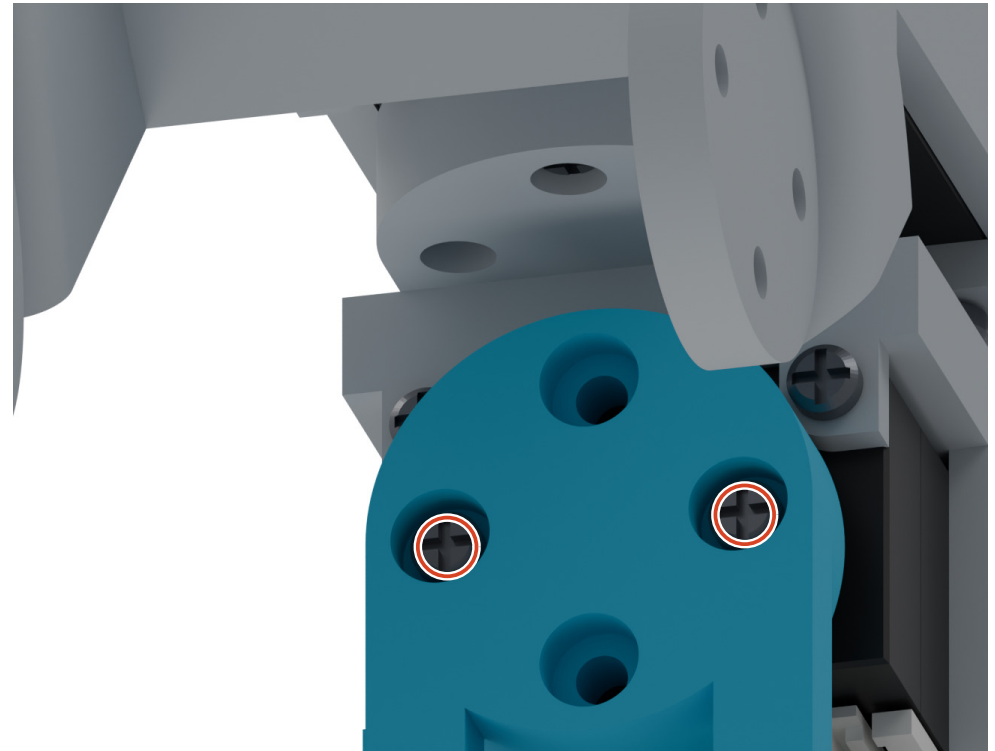
Opposite pairs · 90° apart

35

INSERT – angled view



SEATED – screws fitted

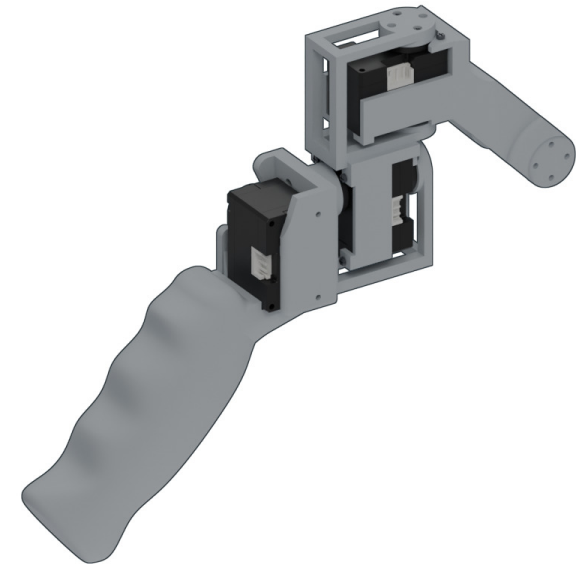
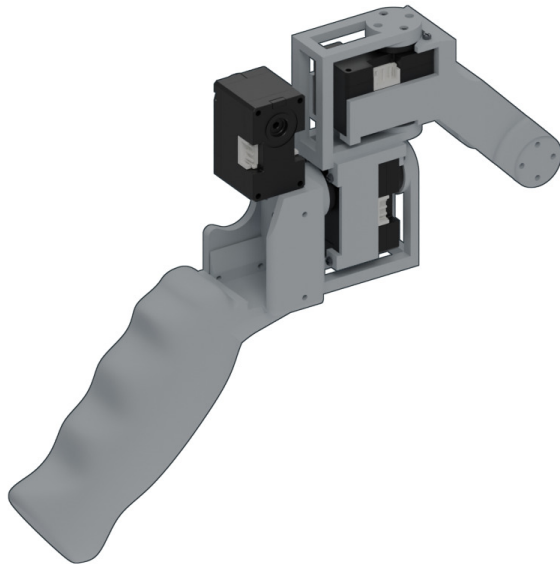


Insert 2 screws to fasten the printed link to M5's rear wheel.

Right: fit only the circled screws in this step. Other screws were fitted earlier.



36



Lower M6 through the open top of its holder in the handle.

M6 has no rear wheel. Wiring follows the mechanical assembly.

Next: insert the screws shown in the close-ups.



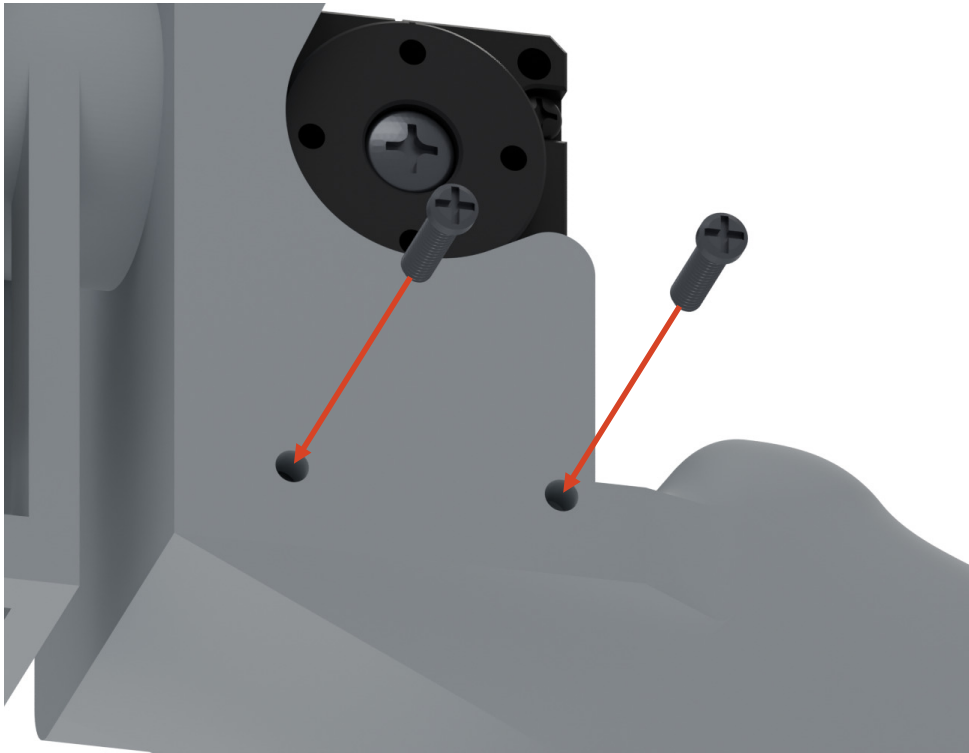
2× M2 × 8 TAP

Use this many on this face

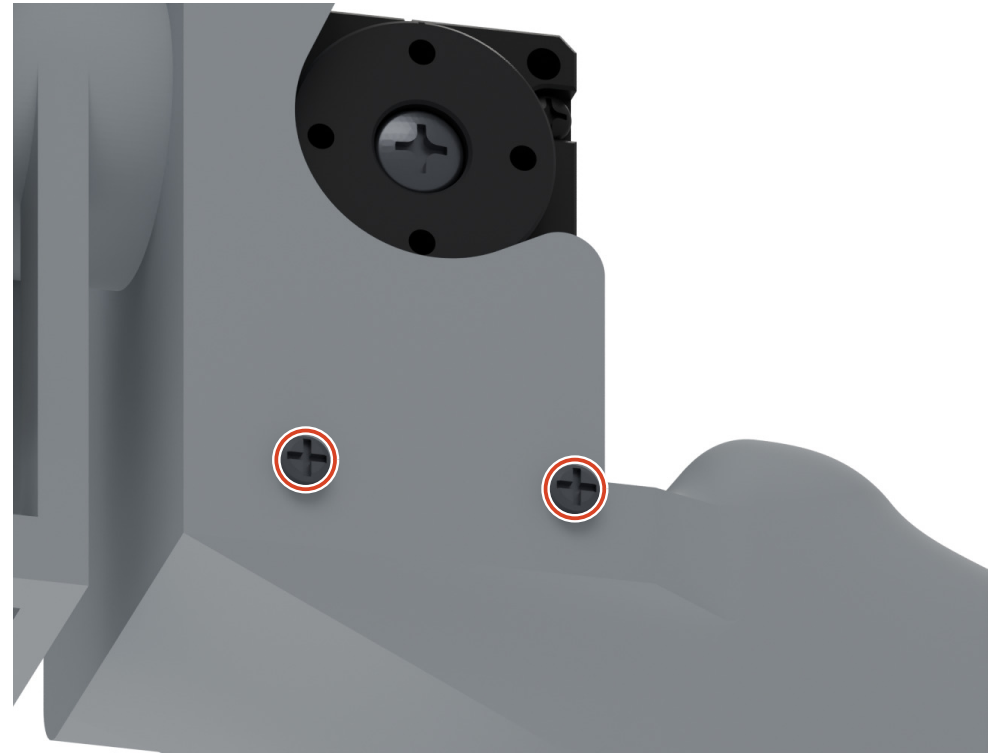
FRONT — horn side

37

INSERT — angled view



SEATED — screws fitted



Insert 2 screws to fasten motor M6 to its printed holder.

Right: fit only the circled screws in this step. Other screws were fitted earlier.



2× M2 × 8 TAP

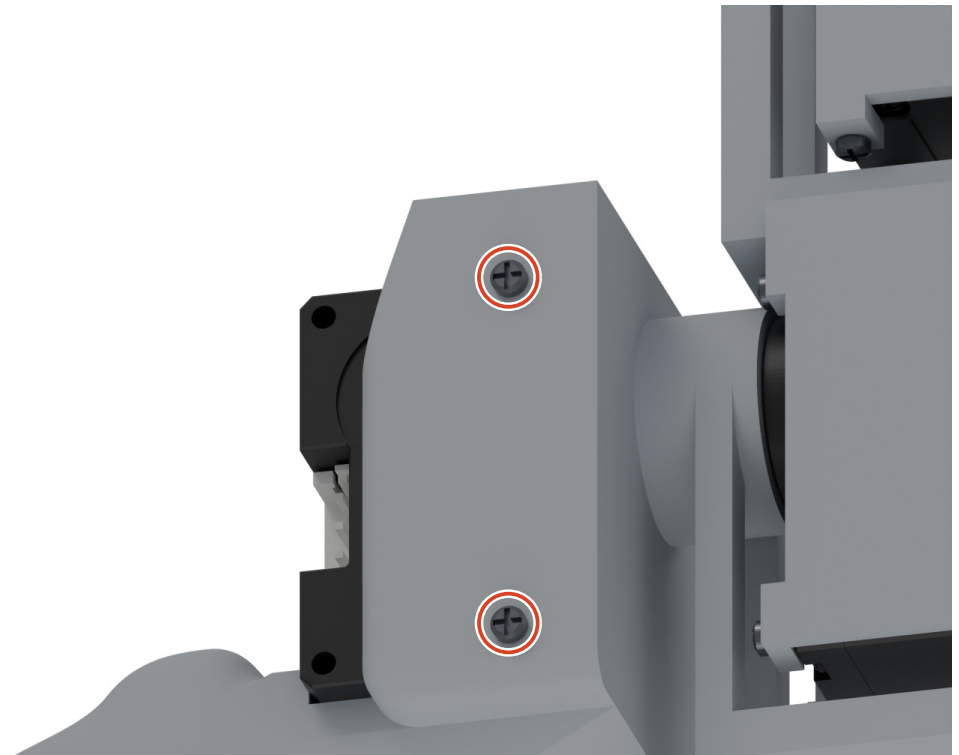
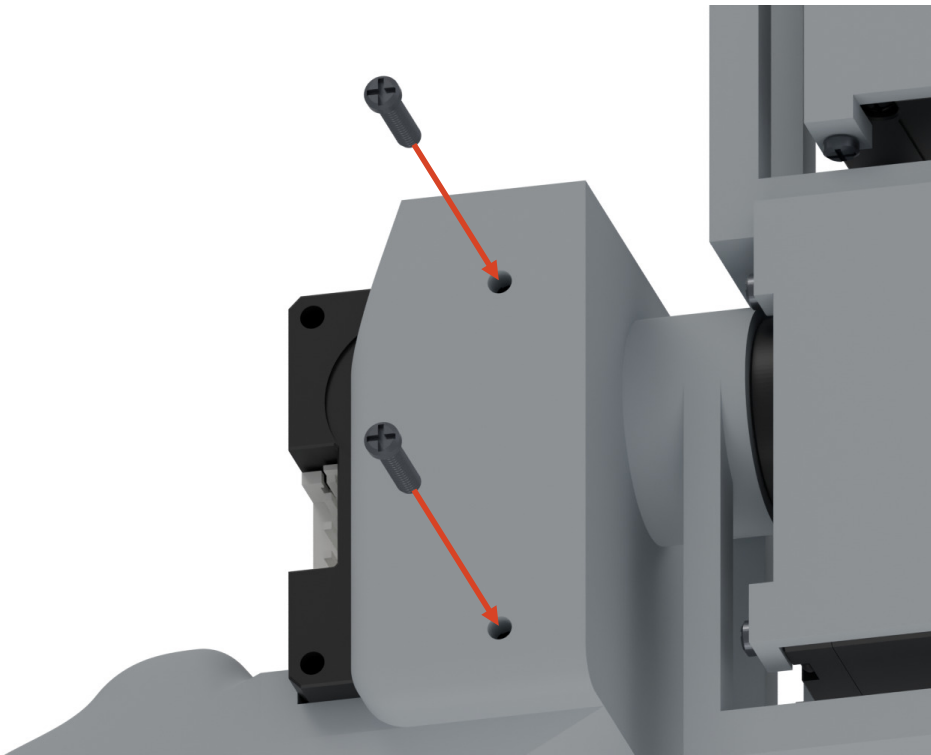
Use this many on this face

REAR — motor body

38

INSERT — angled view

SEATED — screws fitted

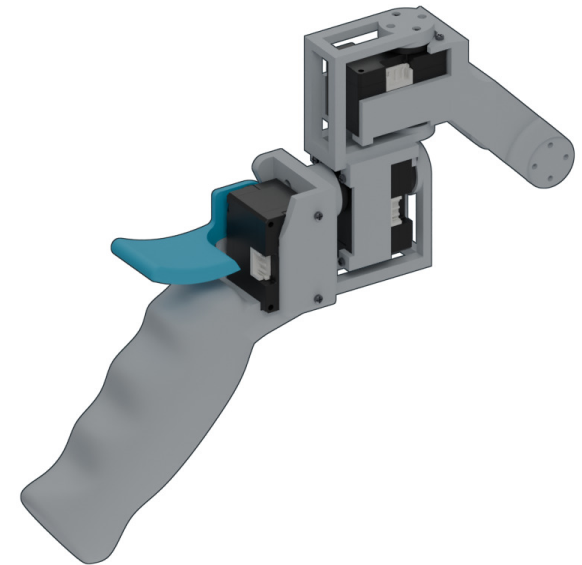
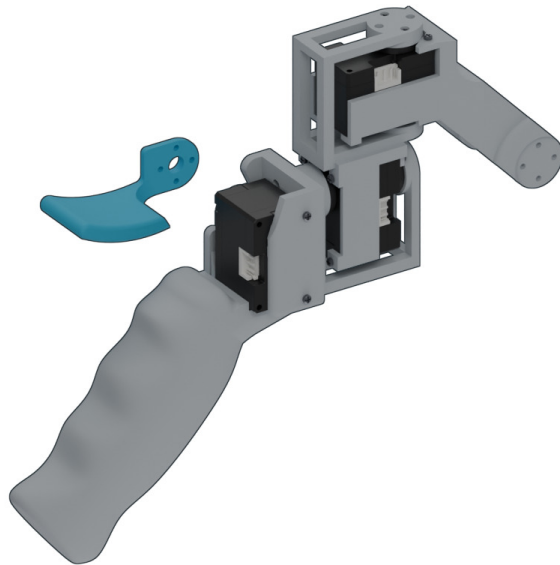


Insert 2 screws to fasten motor M6 to its printed holder.

Right: fit only the circled screws in this step. Other screws were fitted earlier.



39



Attach the trigger to M6's front horn.

Check the trigger's full travel.

Next: insert the screws shown in the close-ups.



4× M2 × 6 TAP

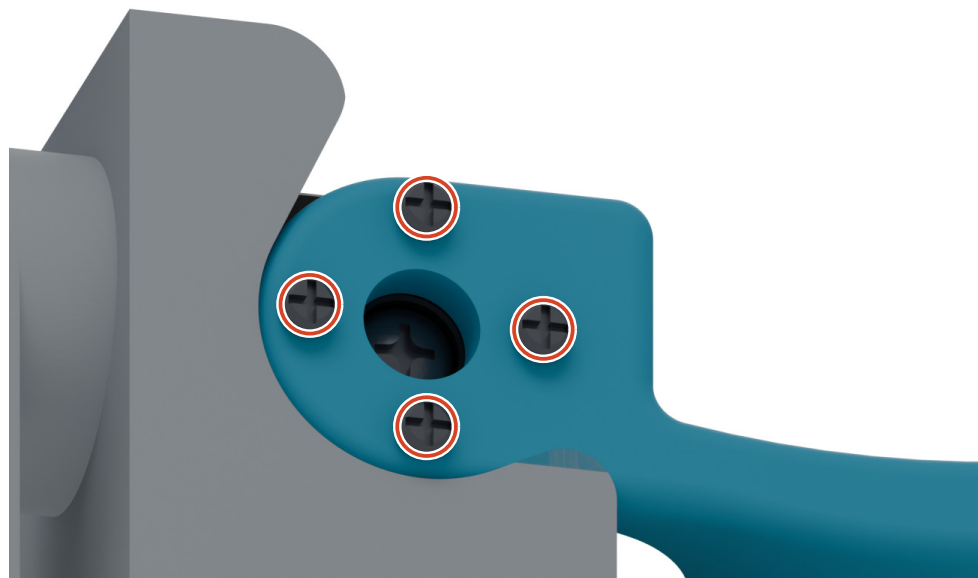
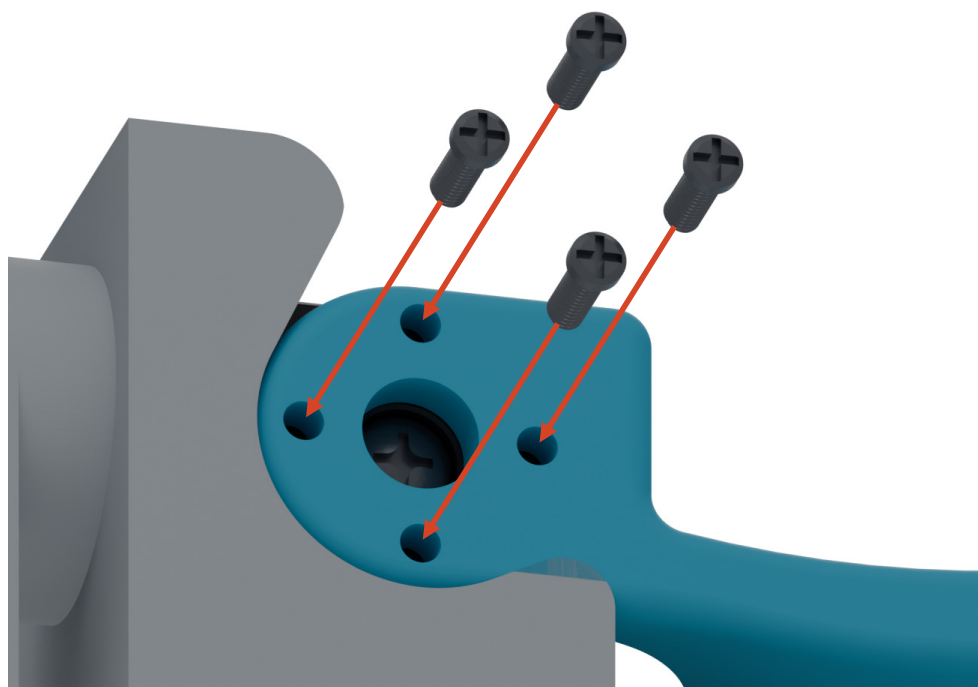
Use this many on this face

FRONT — horn side

40

INSERT — angled view

SEATED — screws fitted

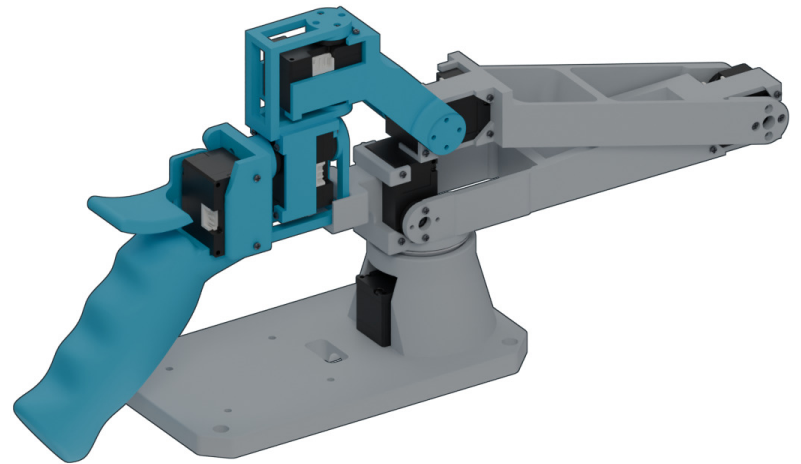
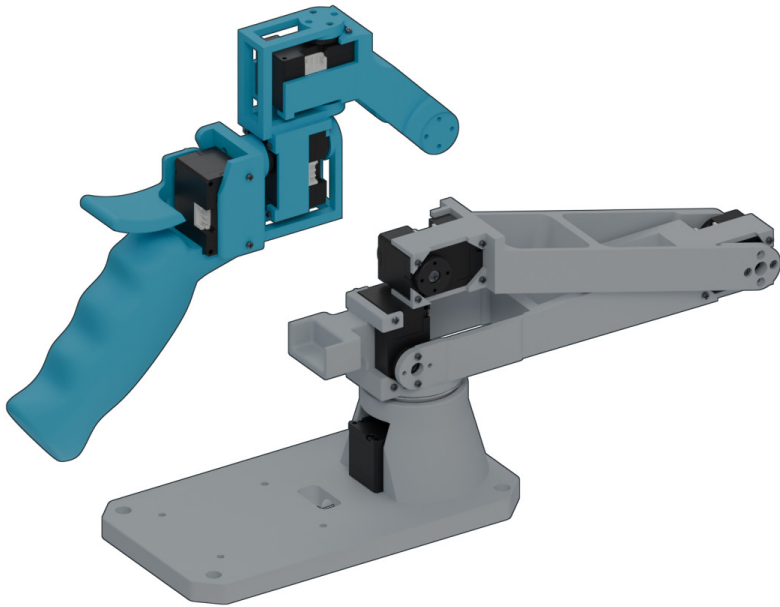


Insert 4 screws to fasten the printed link to M6's front horn.

Right: fit only the circled screws in this step. Other screws were fitted earlier.



41



Lower the completed wrist over M3's horn and rear wheel.

Fit two opposite screws per face; rotate the rear pair 90° from the front pair.

Next: insert the screws shown in the close-ups.



2× M2 × 6 TAP

Use this many on this face

FRONT — horn side



FRONT

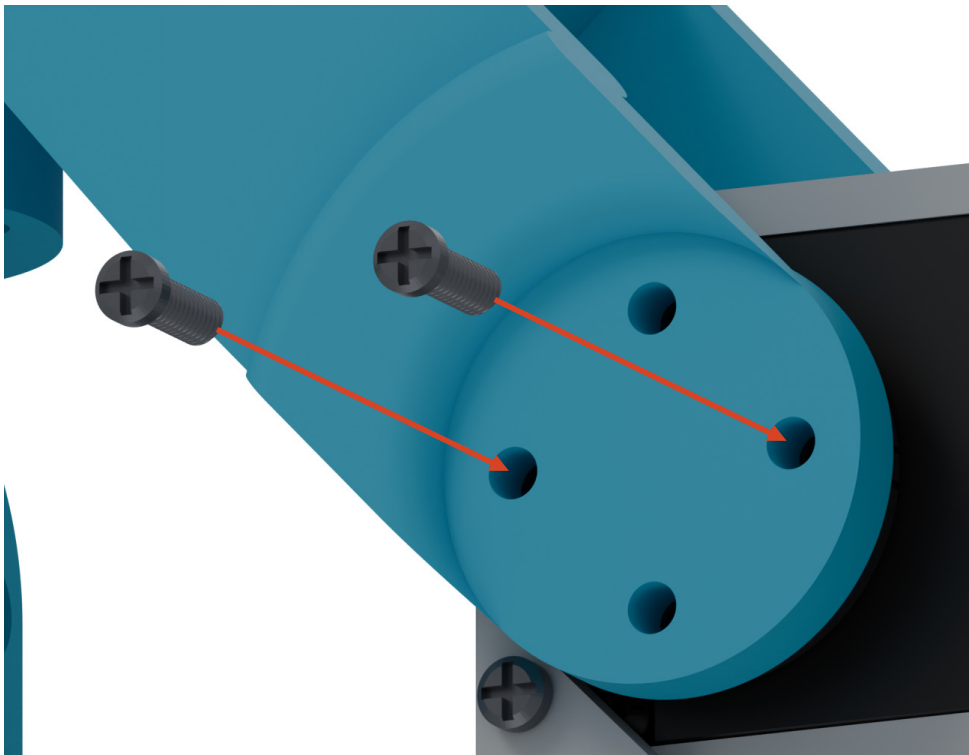


REAR

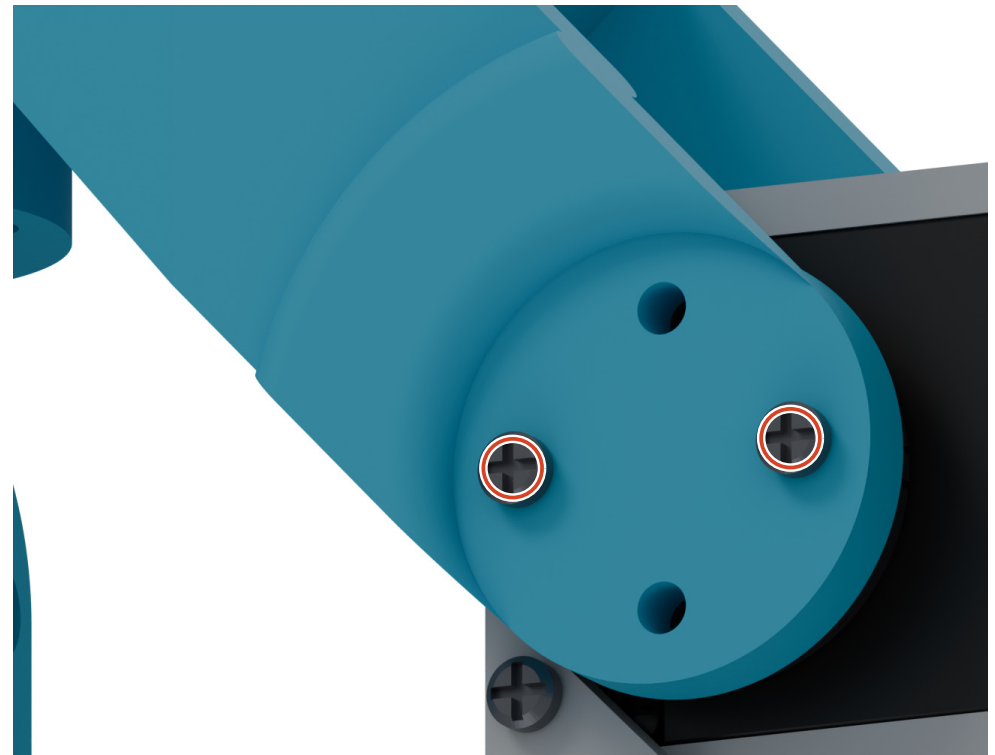
Opposite pairs · 90° apart

42

INSERT — angled view



SEATED — screws fitted



Insert 2 screws to fasten the printed link to M3's front horn.

Right: fit only the circled screws in this step. Other screws were fitted earlier.



2× M2 × 6 TAP

Use this many on this face

REAR – wheel side



FRONT

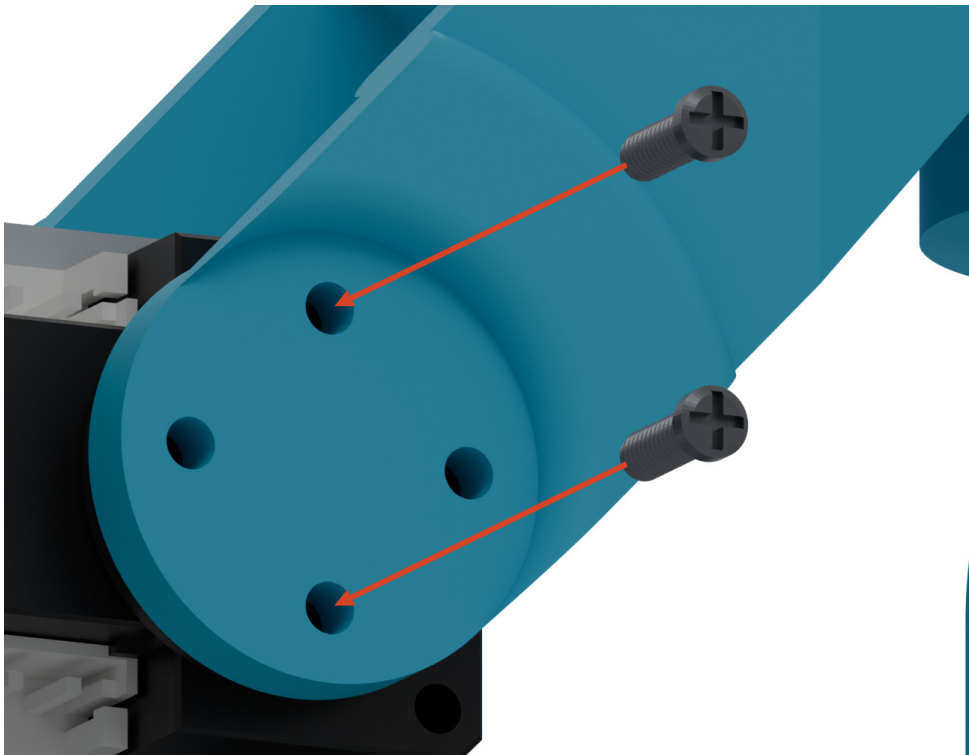


REAR

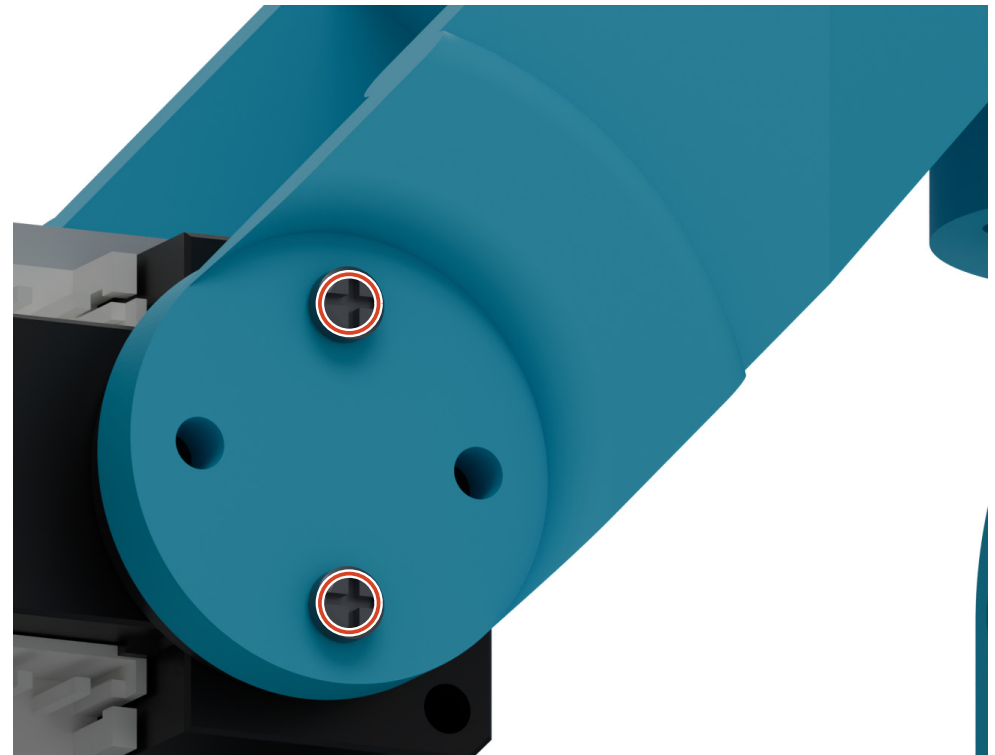
Opposite pairs · 90° apart

43

INSERT – angled view



SEATED – screws fitted

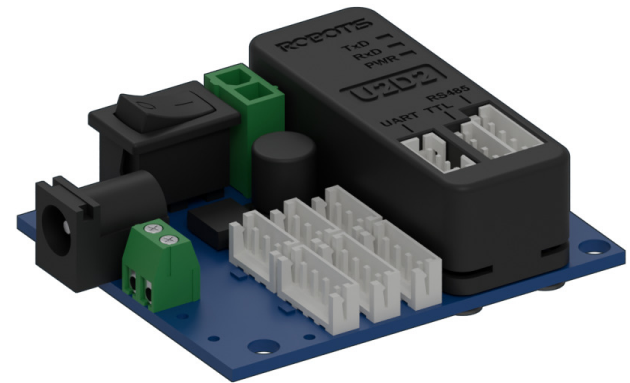
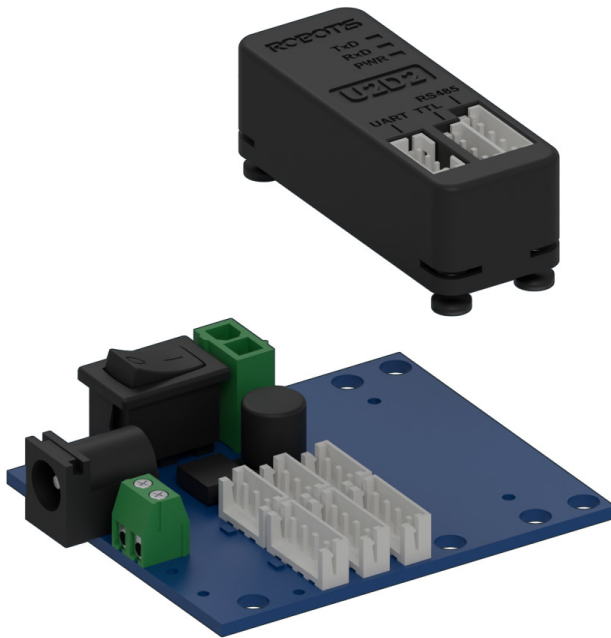


Insert 2 screws to fasten the printed link to M3's rear wheel.

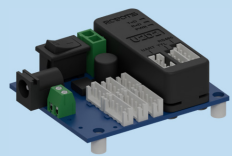
Right: fit only the circled screws in this step. Other screws were fitted earlier.



44

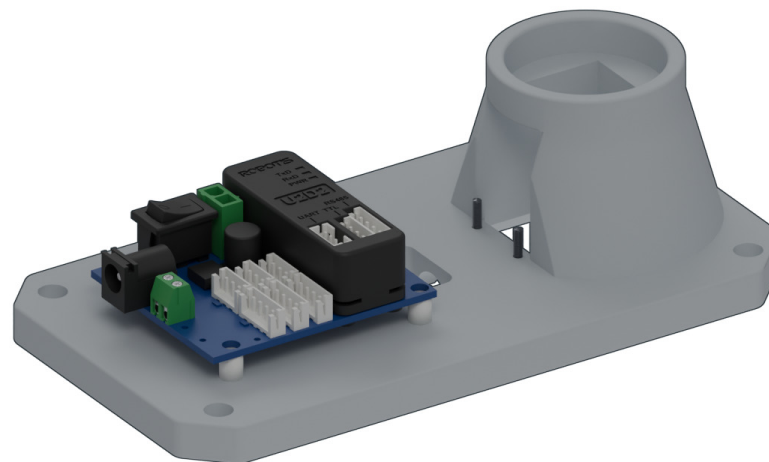
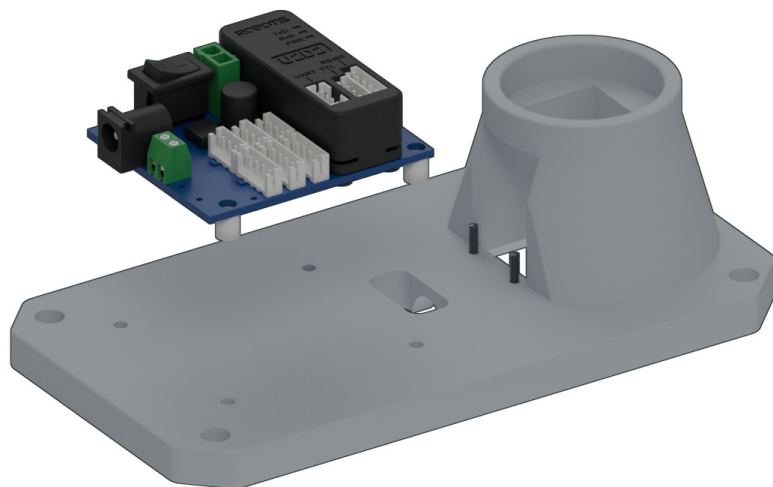


Rivet the U2D2 to the loose hub while its underside is accessible.
Complete the rivets before mounting the hub to the base.



Hub + 4× spacers

45



Attach the completed electronics assembly to the four base mounts.

Use the included power-hub spacers and four M3 × 10 mm screws from the BOM screw kit.

Next: insert the screws shown in the close-ups.



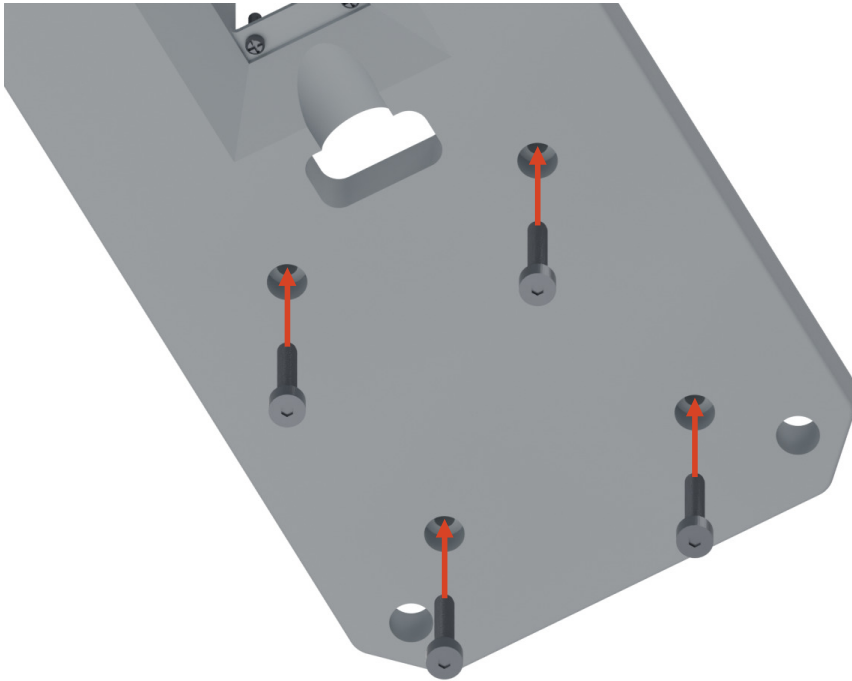
4× M3 × 10 mm

From the BOM screw kit

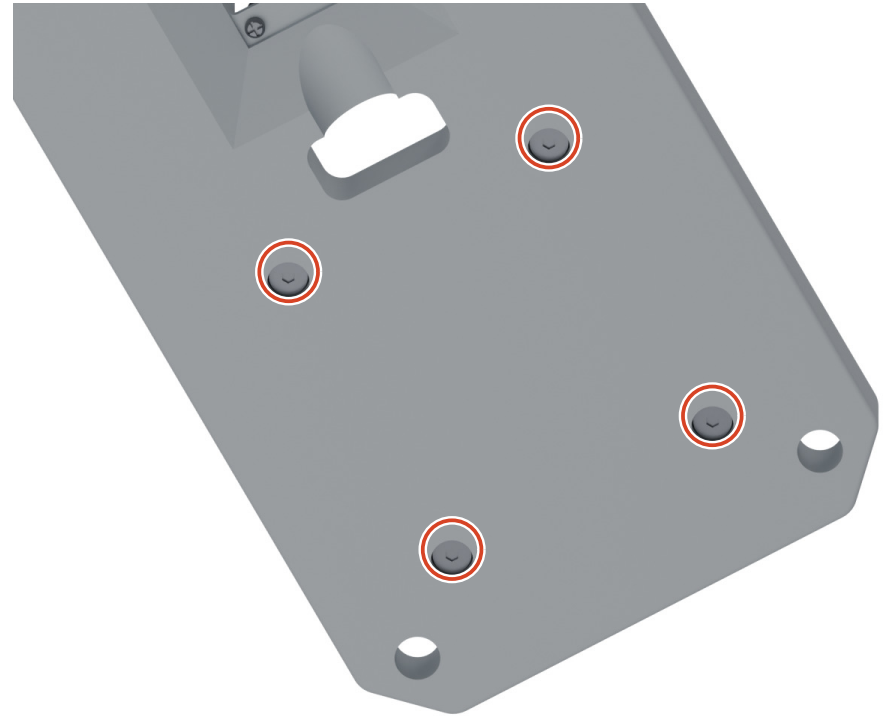
UNDERSIDE

46

INSERT — angled view



SEATED — screws fitted



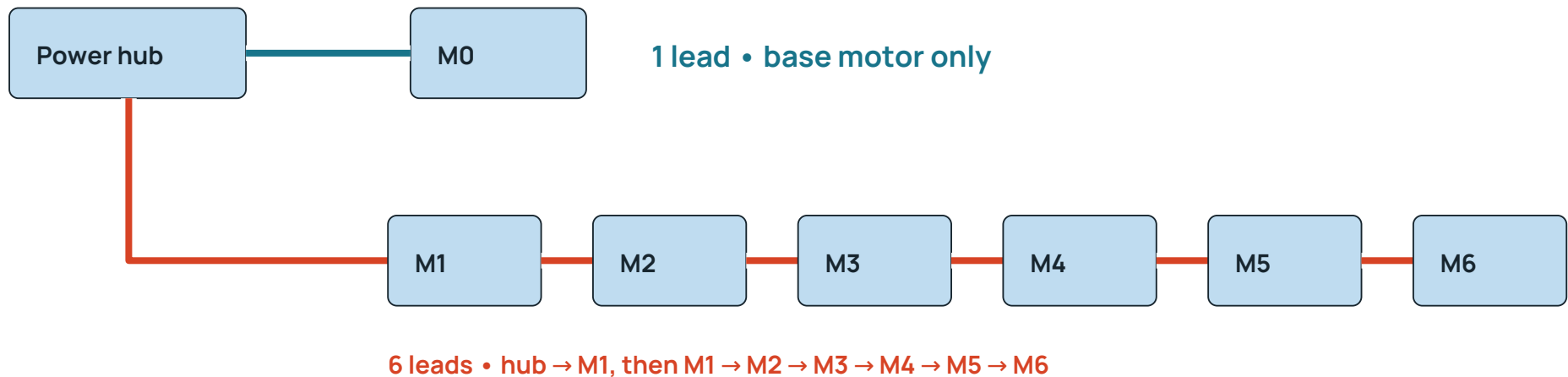
Insert 4 screws to fasten the hub through the base and spacers.

Right: fit only the circled screws in this step. Other screws were fitted earlier.

M3 × 10 mm screws are included in the BOM screw kit. Spacers are included with the power hub.

47 Wiring – two branches

Mechanical assembly is finished. Now prepare the electrical connections.



Do not connect M0 to M1. Both branches share the same power hub.

Before joining the motors

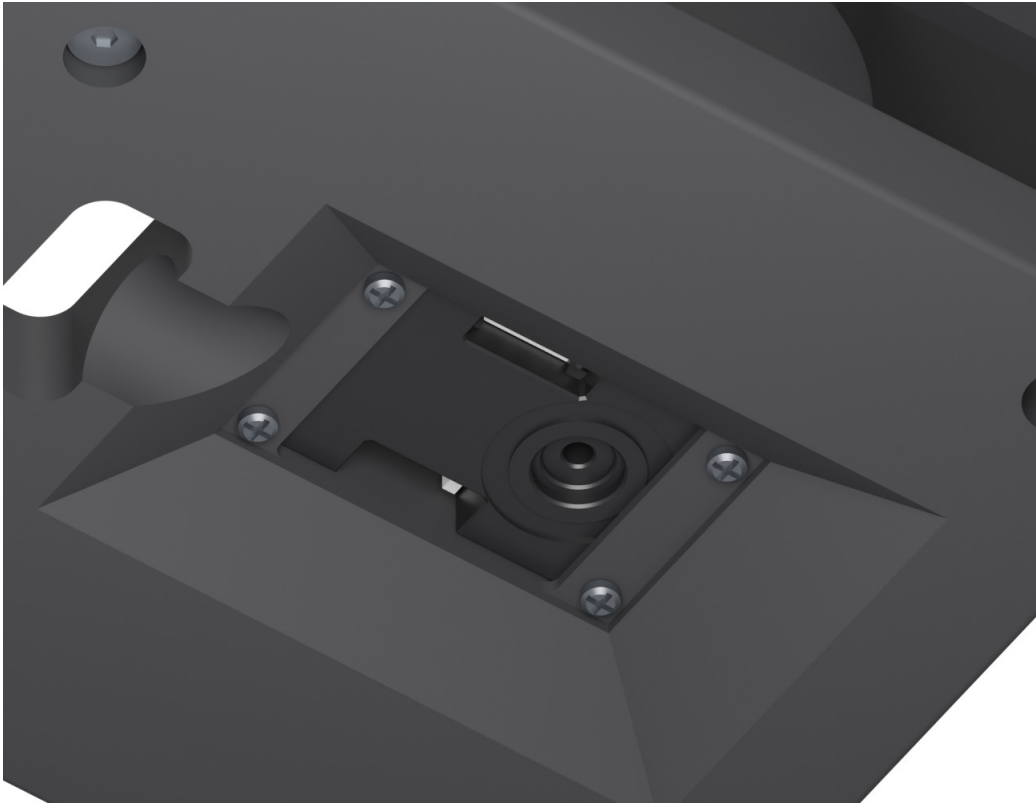
If IDs are not already configured, connect and configure one motor at a time using the controller connection shown in step 50. Match your software's IDs.

M0–M6 identify physical joints; they do not prescribe numeric bus IDs.

Switch power off before connecting or disconnecting any motor lead.

48 Connect the base motor separately

1× 3-pin TTL motor lead, with connector ends that match the hub and XL330

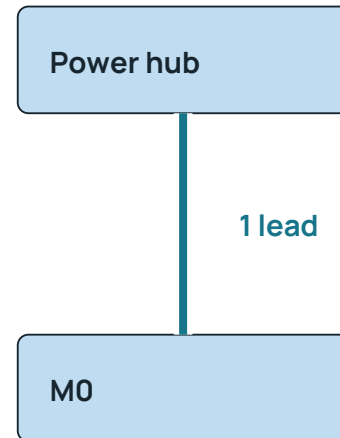


UNDERSIDE • base motor M0

Route the lead through the base's underside tunnel to M0's white TTL socket.

Connect the other end to a 3-pin TTL port on the power hub.

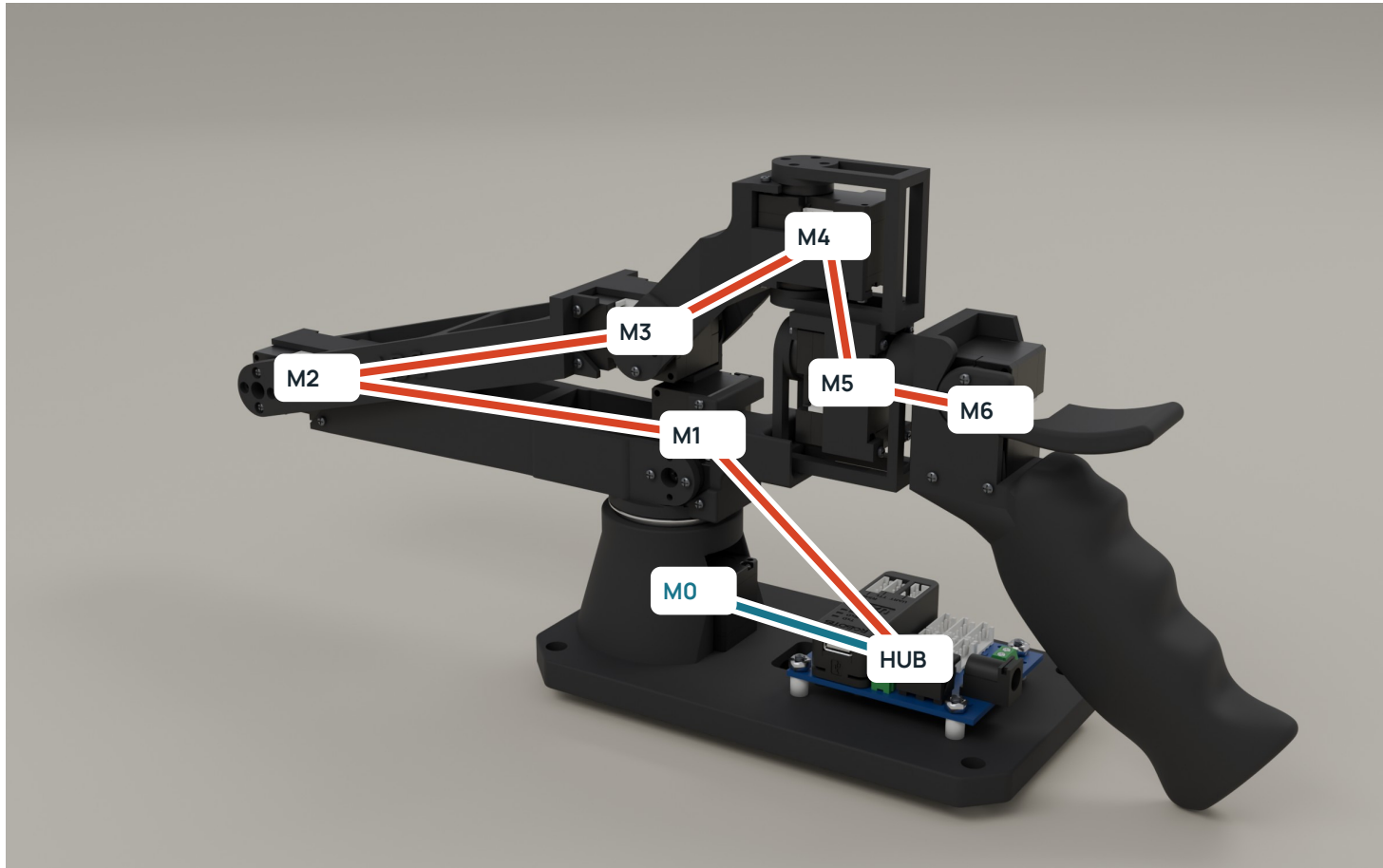
Keep the lead below the bearing and clear of moving parts. Power stays off.



Leave M0's other socket free.

49 Connect the six-motor chain

6× 3-pin TTL leads • one hub lead + five links between motors



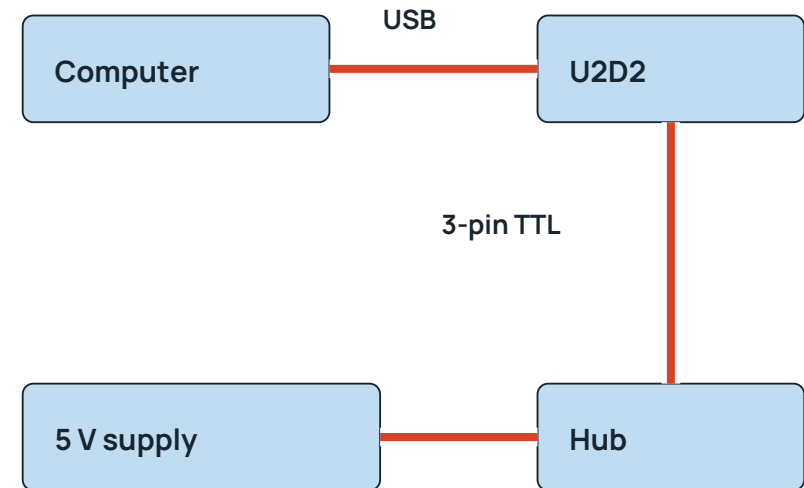
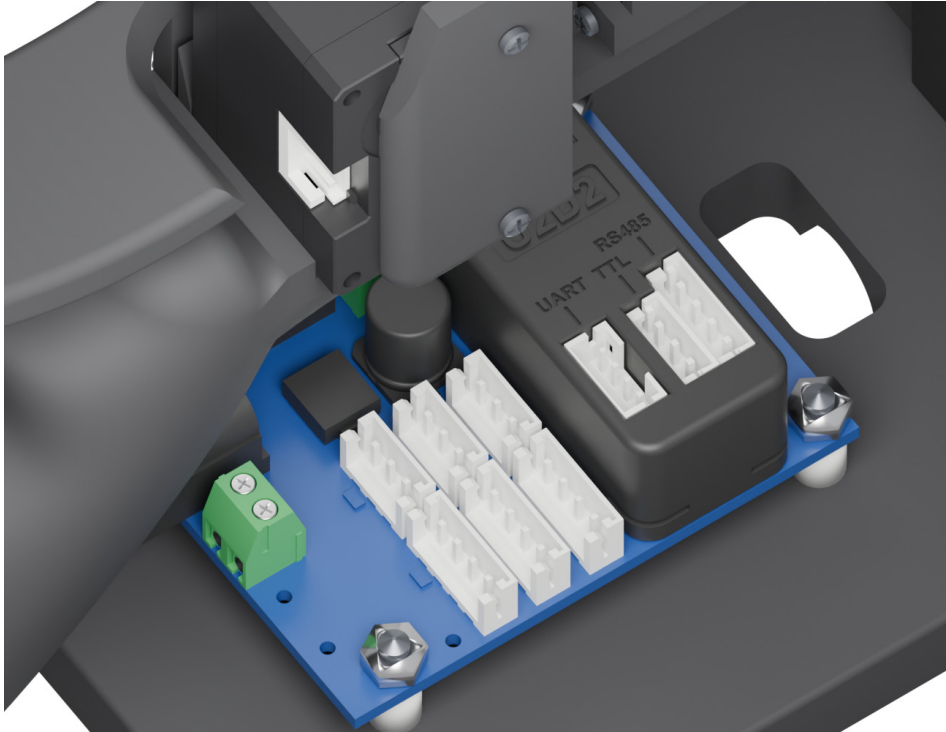
Hub → M1 → M2 → M3 → M4 → M5 → M6

At M1–M5, one socket receives a lead and the other continues to the next motor.

Leave M6's second socket free. Allow a slack loop at every moving joint.

Lines show electrical connections, not the physical cable route. Check full travel by hand.

50 Connect the controller and power

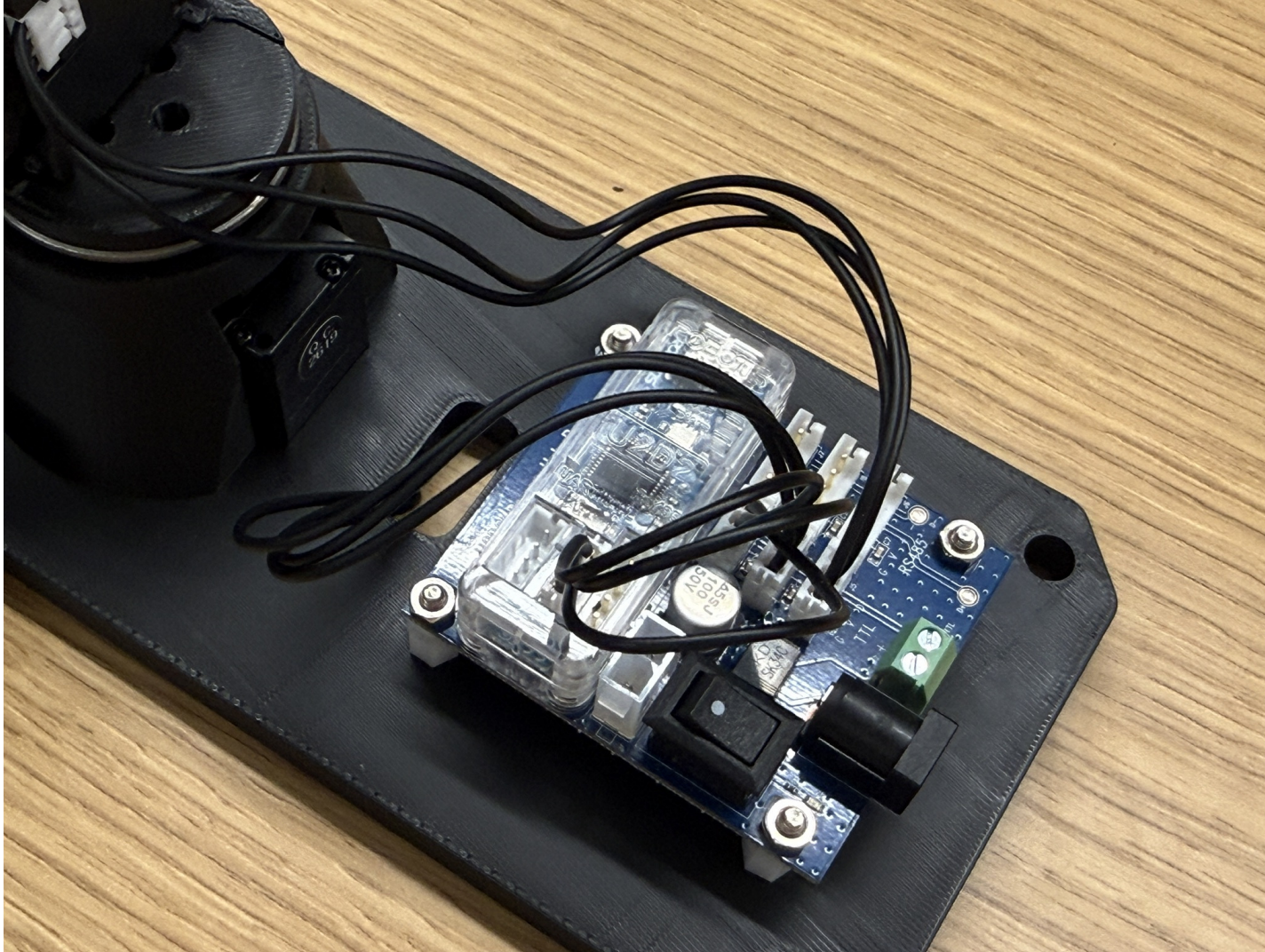


Use the TTL connector, not UART or RS485.

U2D2 on the power hub

1. Connect USB to U2D2, then its 3-pin TTL port to the hub with a matching lead.
2. Connect the 5 V supply to one hub power input. Keep the switch off until ready.
3. Check both motor branches and joint slack, then switch on and check all seven IDs.

Wiring reference – physical build

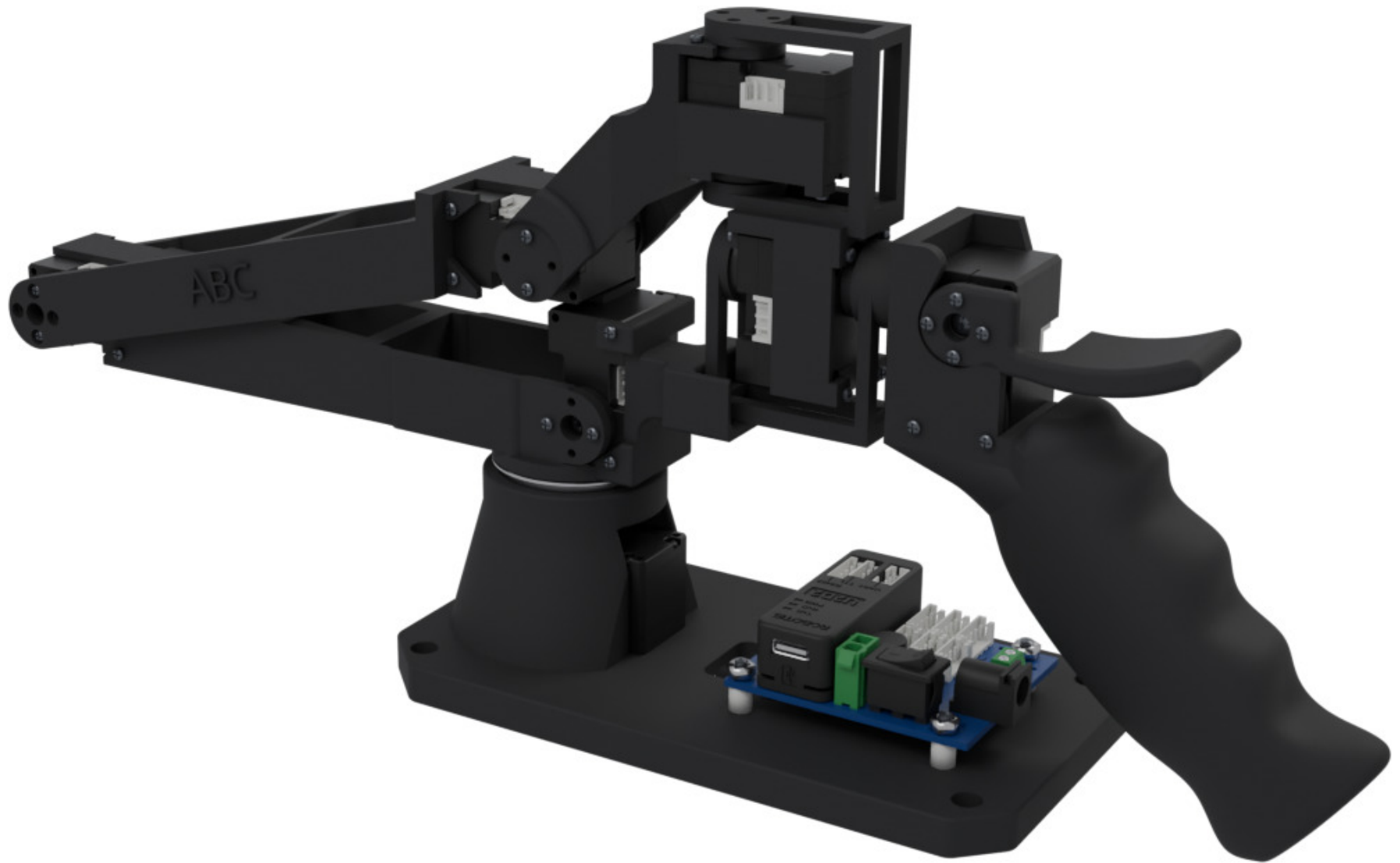


Reference for cable routing around the power hub and through the base opening.

Connections: hub → M0 separately; hub → M1 → M2 → M3 → M4 → M5 → M6.

Some cable ends are obscured in the photo; use the connection diagrams to identify each lead.

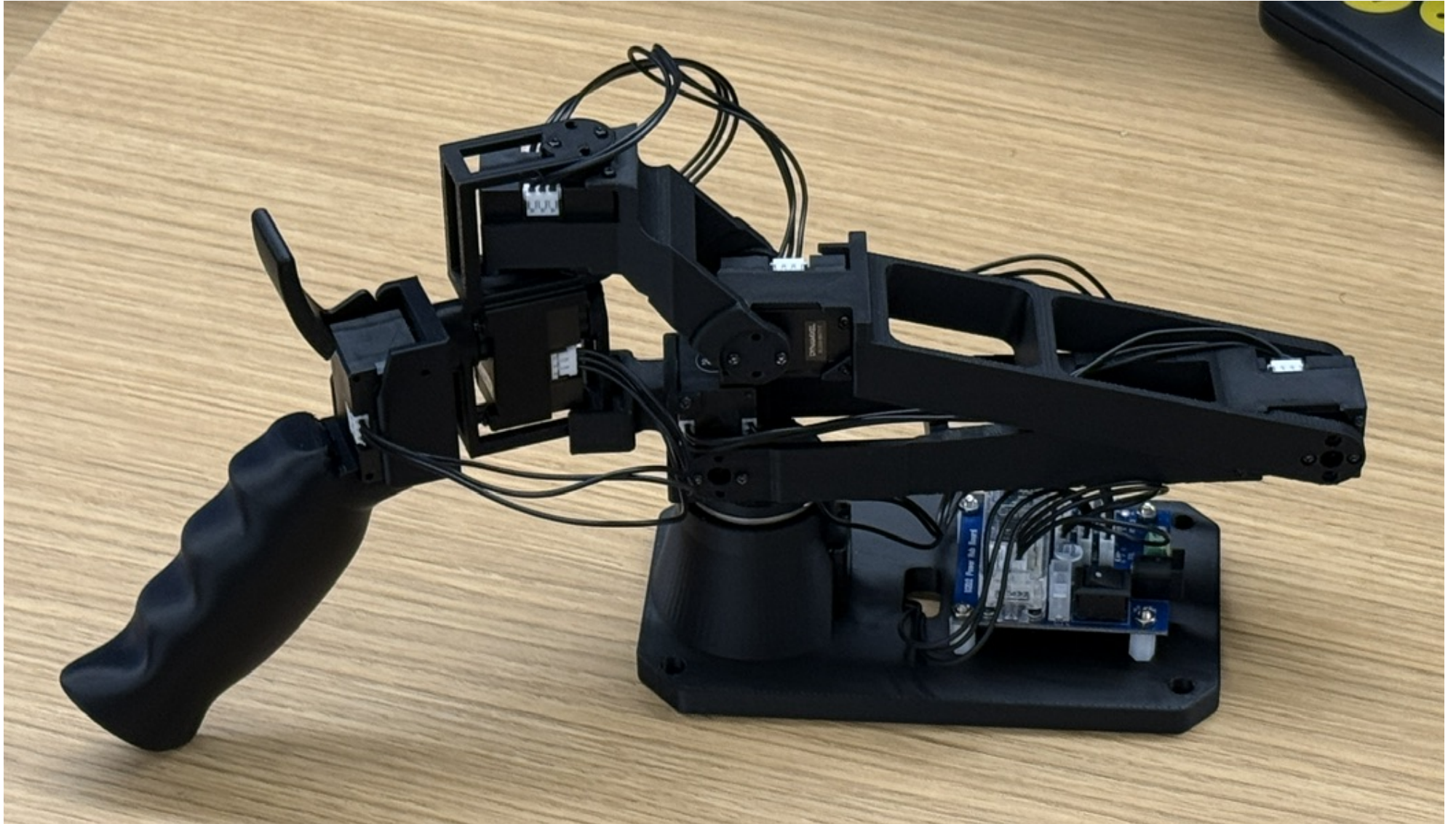
✓ Complete



Check: free joint motion • clear cable loops • secure motor and wheel screws

Counts follow the CAD holes; verify thread engagement and physical tool access during the build.

Completed build



Builder's assembled GELLO – physical build reference